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Acknowledgements

I am very grateful to David for providing me with this template.

But seriously, you don't want to include this sentense, so think about your supervisor, colleagues, family…

Contents

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Chapter 1

Introduction

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Chapter 2

My Papers

1. T. Glechner, P. H. Mayrhofer, D. Holec, S. Fritze, E. Lewin, V. Paneta, D. Primetzhofer, S. Kolozsvári, and H. Riedl, Sci. Rep. **8**, 17669 (2018)
2. S. Veprek, A. S. Argon, and R. F. Zhang, Philos. Mag. **31–32**, 4104 (2010)

Publication I

Pressure-dependent stability of cubic and wurtzite phases within the TiN-–AlN and CrN—AlN systems

D. Holec, F. Rovere, P.H. Mayrhofer, and P.B. Barna

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Pressure-dependent stability of cubic and wurtzite phases within the TiN–AlN and CrN–AlN systems

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Ab initio calculations are employed to demonstrate a strong pressure dependence of the maximum AlN mole fraction preserving the cubic phase in $Ti_{1-x}Al_xN$ and $Cr_{1-x}Al_xN$ hard coatings before transforming to the wurtzite structure. Under a compression of 4 GPa an increase by ~0.1 AlN mole fraction is obtained for both systems. Consequently, stress-related effects cannot be neglected in the discussion of the wide spread of experimentally obtained stability ranges.

Keywords: Ab initio electron theory; Metastable phases; Coatings; Residual stresses; Nitrides

$Ti_{1-x}Al_xN$ and $Cr_{1-x}Al_xN$ hard coatings are widely used in industrial applications due to their superior mechanical and chemical properties such as high hardness, good wear resistance, and high oxidation and corrosion resistance [1–5]. These excellent properties, which in general improve with increasing Al content, are, however, obtained only for the metastable cubic structure (B1, NaCl-like). Al-rich phases prefer the hexagonal (wurtzite B4, ZnS-like) structure which possesses significantly poorer mechanical properties [1,6]. Therefore knowledge of maximum AlN mole fraction in the cubic phase is of basic interest.

There are several ways to calculate the maximum solubility of one phase in another. Makino [7] used a semi-empirical band parameter method to predict the maximum mole fraction of AlN while maintaining the cubic structure as 0.65 in the $Ti_{1-x}Al_xN$ system and 0.77 in the $Cr_{1-x}Al_xN$ system. Zhang and Veprek [6] used a combined ab initio and thermodynamic approach to obtain the maximum solubility of AlN in $Ti_{1-x}Al_xN$ to be 0.68. Mayrhofer et al. [8,9] employed a fully ab initio approach to calculate energies of formation, E_f, of the cubic B1 and wurtzite B4 phases for several discrete compositions. Assuming that E_f depends smoothly on the composition x, the intersection point of E_f^{cubic} and E_f^{wurtzite} determines the solubility limit. Mayrhofer et al. obtained values of ~0.69 for $Ti_{1-x}Al_xN$ [8] and ~0.66 for $Cr_{1-x}Al_xN$ [9].

The experimental observations of the maximum AlN fraction in the cubic phase in $Cr_{1-x}Al_xN$ and $Ti_{1-x}Al_xN$ are not conclusive. The reports for $Ti_{1-x}Al_xN$ span a wide range from 0.4 [10] to 0.68 [11], 0.8 [12] and 0.91 [13]. Similarly, the solubility limit for $Cr_{1-x}Al_xN$ varies in the range 0.6–0.8 [4,14].

It should be noted that the above mentioned ab initio data correspond to 0 K temperatures. The E_f curves (in particular the wurtzite one) are quite flat around the maximum solubility limit, $x_{\max}$. Thus a change of the order 10^1 meV atom^{-1} between E_f^{cubic} and E_f^{wurtzite} corresponds to a shift of a few per cent in $x_{\max}$ [8,9]. Contributions from the vibrational energy at finite temperatures as well as imperfections of the crystal lattices, which differ from material to material and from phase to phase, may account for the discrepancies between experiments and theory. Mayrhofer et al. [8,9] discussed the effect of a partial clustering and showed that, depending on the actual configuration on the atomic level, the cubic–wurtzite E_f intersection can move by as much as 0.1 AlN mole fraction.

Here we discuss an influence of another important parameter, the pressure p. It has been experimentally shown that stresses in thin films strongly depend on the process conditions, in particular, where the energy of the impinging particles is crucial (e.g. by applying higher bias potential). With increasing bias potential

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(which is generally connected with increasing compressive stresses [15–20]), higher maximum mole fractions of AlN in the cubic phase of $Ti_{1-x}Al_xN$ based materials are obtained [21]. The stresses in the experiments are usually biaxial in character. In this study, however, we consider the hydrostatic pressure for which additional information such as the individual elastic constants, etc., is not needed. It should be noted that positive values of pressure correspond to compression.

The internal energy, U, is the correct thermodynamic potential at zero pressure and zero temperature. The formation energy per atom, E_f, is defined (e.g. for the $Ti_{1-x}Al_xN$ system) as follows:

$$E_f^{\xi} = E^{\xi}_{\mathrm{Ti}_{1-x}\mathrm{Al}_x\mathrm{N}} - \frac{1}{2}((1-x)E_{\mathrm{Ti}} + xE_{\mathrm{Al}} + E_{\mathrm{N}}). \tag{1}$$

Here ξ corresponds to either cubic or wurtzite phase, and E_X is the single element total energy corresponding to the internal energy of pure X (crystalline in the case of Al and Ti, or gas molecules in the case of N_2). Consequently, the maximum solubility limit, which is given as $E_f^{\mathrm{cubic}}(x_{\max}) = E_f^{\mathrm{wurtzite}}(x_{\max})$, is equivalent to $U^{\mathrm{cubic}}(x_{\max}) = U^{\mathrm{wurtzite}}(x_{\max})$. When the pressure is considered (still at zero temperature), the enthalpy, $H = U + pV$, must be taken as the correct thermodynamic potential instead of the internal energy U. The Murnaghan equation of state [22]:

$$E(V) = E_0 + \frac{B_0V}{B_0'}\left(\frac{(V_0/V)^{B_0'}}{B_0'-1} + 1\right) - \frac{B_0V_0}{B_0'-1} \tag{2}$$

provides an expression for the equilibrium volume as a function of pressure:

$$p = \frac{\partial E}{\partial V} \Rightarrow V(p) = \frac{V_0}{\sqrt[B_0']{\frac{B_0'}{B_0}p + 1}}. \tag{3}$$

Inserting Eq. (3) into Eq. (2) yields:

$$E(p) = E_0 + \frac{B_0V_0}{B_0'-1}\left(\frac{\frac{p}{B_0} + 1}{\sqrt[B_0']{\frac{B_0'}{B_0}p + 1}} - 1\right). \tag{4}$$

Combining these results together, an analytical expression for the enthalpy as a function of pressure is obtained:

$$H(p) = E_0 - \frac{B_0V_0}{B_0'-1} + \frac{pV_0}{\sqrt[B_0']{\frac{B_0'}{B_0}p + 1}}\left(\frac{\frac{B_0}{p} + 1}{B_0'-1} + 1\right). \tag{5}$$

Since this expression contains the same parameters as the Murnaghan equation of state (Eq. (2)), its quality can be controlled/improved by, for example, calculating additional $E(V)$ data points.

We used the Vienna Ab Initio Simulation Package [23,24] together with the PAW pseudopotentials [25]. The calculation parameters were sufficient to guarantee a precision of several meV atom^{-1} (for details see Refs. [8,9,26]). While the $Ti_{1-x}Al_xN$ system is treated as nonmagnetic, the $Cr_{1-x}Al_xN$ data corresponds to the antiferromagnetic (AFM) arrangement [9]. Although CrN crystallizes in the paramagnetic (PM) structure above the Néel temperature (273–283 K) [27], Alling et al. [28] showed that there is no significant difference in energy of formation and other properties between the PM and the AFM state which is the preferred one under the Néel temperature. If the magnetic moments are not considered, the results vary widely from values obtained by experiment [9].

The datapoints in Figure 1a correspond to a collection of several specially designed cells containing 32 and 64 atoms [8]. In addition, new data points for both cubic and wurtzite phase were calculated using special quasi-random structures (SQS) [29]. The third-order polynomials were fitted to the data at a given pressure in order to estimate the crossover of the cubic and wurtzite phase enthalpies. By adding 4 GPa of compressive pressure to the $Ti_{1-x}Al_xN$ system, the crossover between the cubic and hexagonal phases (i.e. the maximum solubility of AlN, $x_{\max}$, in the cubic phase) increases from 0.70 to 0.78 – compare Figure 1a and b, respectively. Consequently, the cubic phase stability increases by ~11%. It is worth noting that compressive stresses around 4 GPa are commonly present in deposited thin

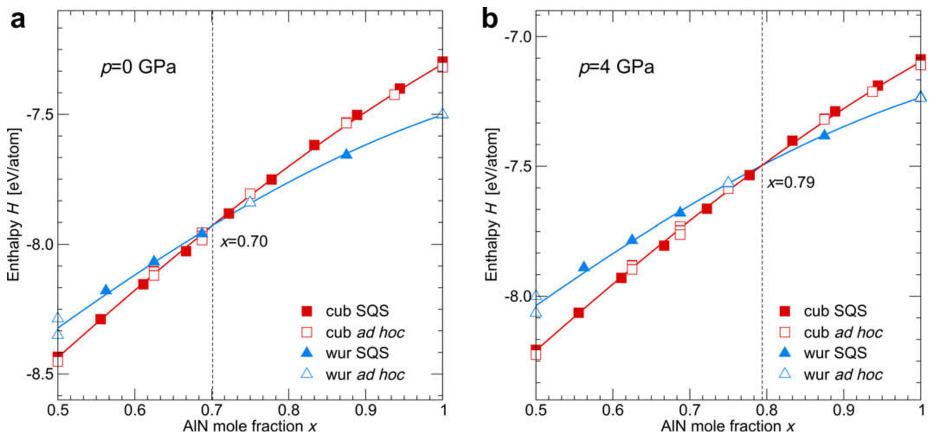

Figure 1. Enthalpies of the cubic (red squares) and wurtzite (blue triangles) phases of the $Ti_{1-x}Al_xN$ system (a) in equilibrium and (b) under a load of 4 GPa hydrostatic pressure. Full and open symbols represent the SQS and the ad hoc configurations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this paper.)

films [20] and thus this model consideration is based on a realistic situation. The dependence of x_{max} in $Ti_{1-x}Al_xN$ as a function of pressure is shown in Figure 2. The trend for increasing the solubility limit with applying larger compressive hydrostatic pressures is predictable since AlN transforms from the wurtzite to the cubic phase at ∼16 GPa (a value based on both experiments and calculations) [30,31]. Since the cubic phase has a smaller volume per atom than the wurtzite phase [6], TiN maintains its cubic structure also under compression. Consequently, one would expect that above such pressure the cubic phase is stable for the whole compositional range: indeed, for ∼14 GPa the enthalpies of the cubic and wurtzite phases crosses at $x_{max} = 1$, which is in agreement with the value for AlN [30,31].

Subsequently, we applied the same methodology to the $Cr_{1-x}Al_xN$ system. This system is more difficult when compared with $Ti_{1-x}Al_xN$ since the proper magnetic state of the Cr atoms must be taken into account. The difficulties with describing magnetic random alloys are responsible for the absolute values of the ab initio calculated x_{max} to be less accurate than in the case of $Ti_{1-x}Al_xN$.

The data of Mayrhofer et al. [9] used for the phase-stability calculation is based on ad hoc cells. To strengthen our analysis we the added SQS-based data-points of Rovere et al. [26]. In agreement with the $Ti_{1-x}Al_xN$ system, as for the $Cr_{1-x}Al_xN$ system the ad hoc data points fall into the vicinity of the SQS cells (see Figure 3a). The addition of compression increases the maximum AlN solubility of the cubic phase also in the $Cr_{1-x}Al_xN$ system. Under a compressive pressure of 4 GPa, x_{max} is around 0.79 (see Figure 3b) which is ∼20% higher than x_{max} of ∼0.66 for 0 GPa (see Figure 3a). The calculated maximum solubility of AlN in $Cr_{1-x}Al_xN$ as a function of pressure p is shown in Figure 4. As for the $Ti_{1-x}Al_xN$

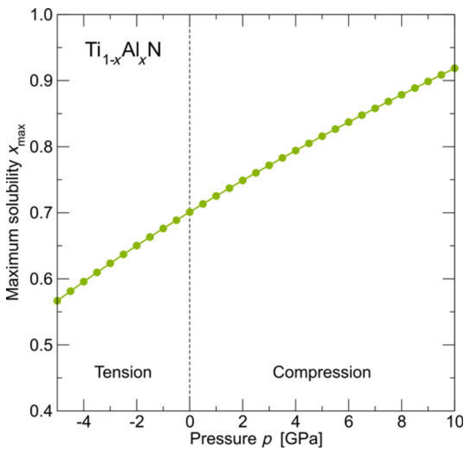

Figure 2. Calculated maximum mole fraction of AlN in the cubic B1 phase of the $Ti_{1-x}Al_xN$ system.

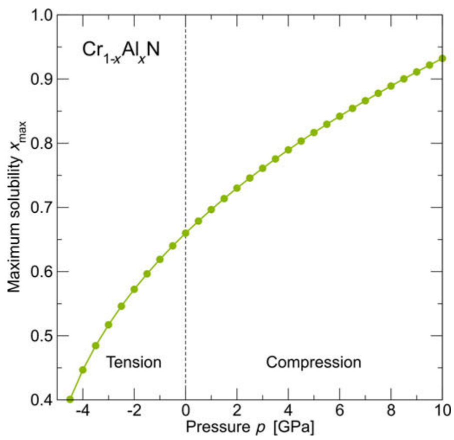

Figure 4. Calculated maximum mole fraction of AlN in the cubic B1 phase of the $Cr_{1-x}Al_xN$ system.

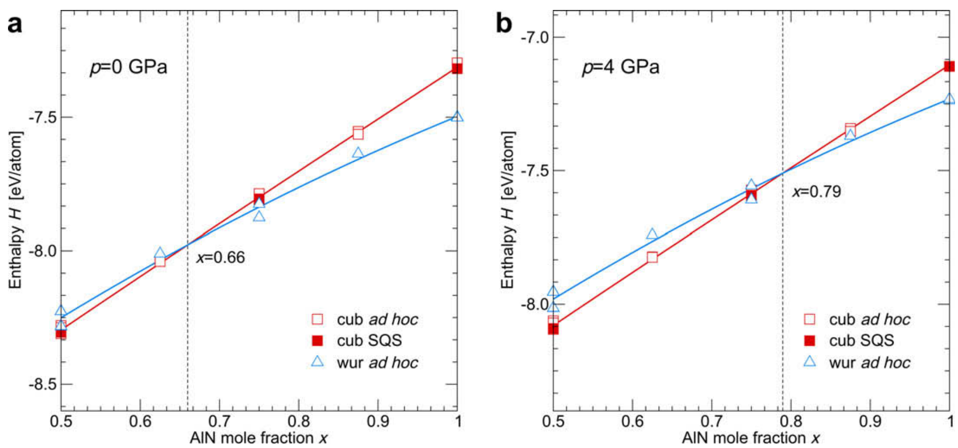

Figure 3. Enthalpies of the cubic (red squares) and wurtzite (blue triangles) phases of the $Cr_{1-x}Al_xN$ system (a) in equilibrium and (b) under a load of 4 GPa hydrostatic pressure. Full and open symbols represent the SQS and the ad hoc configurations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this paper.)

system, x_{max} increases with increasing compressive pressure. The dependence is steeper for $Cr_{1-x}Al_xN$ than in the case of $Ti_{1-x}Al_xN$, especially in the range of tensile hydrostatic pressures. Consequently, for tensile stresses in the coatings x_{max} drops rapidly below 0.5.

In conclusion, we have studied the effect of tensile and compressive hydrostatic pressure on the solubility limit x_{max} of AlN in the cubic B1 phase of $Ti_{1-x}Al_xN$ and $Cr_{1-x}Al_xN$ systems. It follows that x_{max} depends strongly on the applied pressure; compression of about 4 GPa increases x_{max} by $\sim$0.1 AlN mole fraction for both systems. Although the studied stress state differs from the biaxial stress obtained experimentally in thin films during deposition, we can conclude that the built-in stresses influence the phase stabilities of these systems significantly.

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Publication II

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The internal energy, U, is the correct thermodynamic potential at zero pressure and zero temperature. The formation energy per atom, E_f, is defined (e.g. for the $Ti_{1-x}Al_xN$ system) as follows:

$$E_f^{\xi} = E^{\xi}_{\mathrm{Ti}_{1-x}\mathrm{Al}_x\mathrm{N}} - \frac{1}{2}((1-x)E_{\mathrm{Ti}} + xE_{\mathrm{Al}} + E_{\mathrm{N}}). \quad (1)$$

Here ξ corresponds to either cubic or wurtzite phase, and E_X is the single element total energy corresponding to the internal energy of pure X (crystalline in the case of Al and Ti, or gas molecules in the case of N_2). Consequently, the maximum solubility limit, which is given as $E_f^{\mathrm{cubic}}(x_{\max}) = E_f^{\mathrm{wurtzite}}(x_{\max})$, is equivalent to $U^{\mathrm{cubic}}(x_{\max}) = U^{\mathrm{wurtzite}}(x_{\max})$. When the pressure is considered (still at zero temperature), the enthalpy, $H = U + pV$, must be taken as the correct thermodynamic potential instead of the internal energy U. The Murnaghan equation of state [22]:

$$E(V) = E_0 + \frac{B_0 V}{B_0'}\left(\frac{(V_0/V)^{B_0'}}{B_0'-1} + 1\right) - \frac{B_0 V_0}{B_0'-1} \quad (2)$$

provides an expression for the equilibrium volume as a function of pressure:

$$p = \frac{\partial E}{\partial V} \Rightarrow V(p) = \frac{V_0}{\sqrt[B_0']{\frac{B_0'}{B_0}p + 1}}. \quad (3)$$

Inserting Eq. (3) into Eq. (2) yields:

$$E(p) = E_0 + \frac{B_0 V_0}{B_0'-1}\left(\frac{\frac{p}{B_0}+1}{\sqrt[B_0']{\frac{B_0'}{B_0}p+1}} - 1\right). \quad (4)$$

Combining these results together, an analytical expression for the enthalpy as a function of pressure is obtained:

$$H(p) = E_0 - \frac{B_0 V_0}{B_0'-1} + \frac{pV_0}{\sqrt[B_0']{\frac{B_0'}{B_0}p+1}}\left(\frac{\frac{B_0}{p}+1}{B_0'-1} + 1\right). \quad (5)$$

Since this expression contains the same parameters as the Murnaghan equation of state (Eq. (2)), its quality can be controlled/improved by, for example, calculating additional $E(V)$ data points.

We used the Vienna Ab Initio Simulation Package [23,24] together with the PAW pseudopotentials [25]. The calculation parameters were sufficient to guarantee a precision of several meV atom^{-1} (for details see Refs. [8,9,26]). While the $Ti_{1-x}Al_xN$ system is treated as non-magnetic, the $Cr_{1-x}Al_xN$ data corresponds to the antiferromagnetic (AFM) arrangement [9]. Although CrN crystallizes in the paramagnetic (PM) structure above the Néel temperature (273–283 K) [27], Alling et al. [28] showed that there is no significant difference in energy of formation and other properties between the PM and the AFM state which is the preferred one under the Néel temperature. If the magnetic moments are not considered, the results vary widely from values obtained by experiment [9].

The datapoints in Figure 1a correspond to a collection of several specially designed cells containing 32 and 64 atoms [8]. In addition, new data points for both cubic and wurtzite phase were calculated using special quasi-random structures (SQS) [29]. The third-order polynomials were fitted to the data at a given pressure in order to estimate the crossover of the cubic and wurtzite phase enthalpies. By adding 4 GPa of compressive pressure to the $Ti_{1-x}Al_xN$ system, the crossover between the cubic and hexagonal phases (i.e. the maximum solubility of AlN, $x_{\max}$, in the cubic phase) increases from 0.70 to 0.78 – compare Figure 1a and b, respectively. Consequently, the cubic phase stability increases by ~11%. It is worth noting that compressive stresses around 4 GPa are commonly present in deposited thin

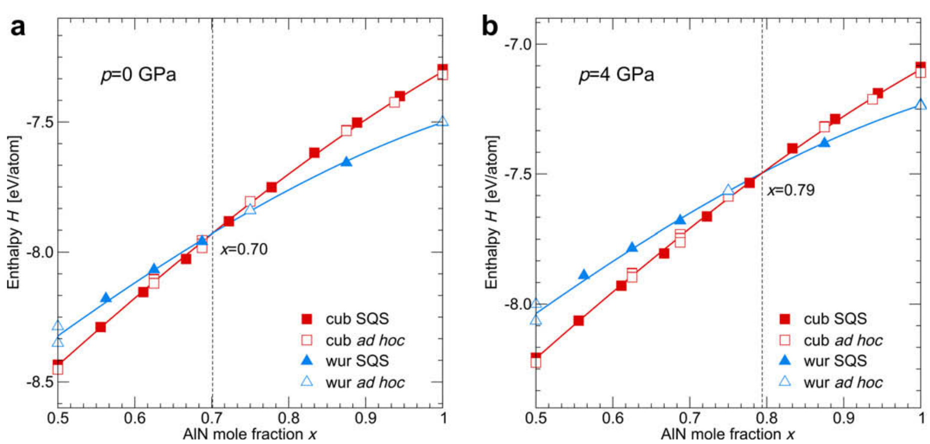

Figure 1. Enthalpies of the cubic (red squares) and wurtzite (blue triangles) phases of the $Ti_{1-x}Al_xN$ system (a) in equilibrium and (b) under a load of 4 GPa hydrostatic pressure. Full and open symbols represent the SQS and the ad hoc configurations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this paper.)

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Subsequently, we applied the same methodology to the $Cr_{1-x}Al_xN$ system. This system is more difficult when compared with $Ti_{1-x}Al_xN$ since the proper magnetic state of the Cr atoms must be taken into account. The difficulties with describing magnetic random alloys are responsible for the absolute values of the ab initio calculated x_{max} to be less accurate than in the case of $Ti_{1-x}Al_xN$.

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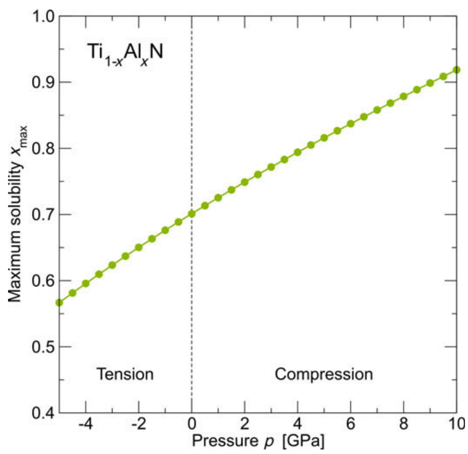

Figure 2. Calculated maximum mole fraction of AlN in the cubic B1 phase of the $Ti_{1-x}Al_xN$ system.

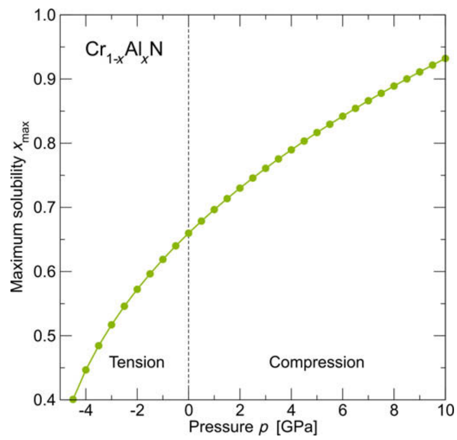

Figure 4. Calculated maximum mole fraction of AlN in the cubic B1 phase of the $Cr_{1-x}Al_xN$ system.

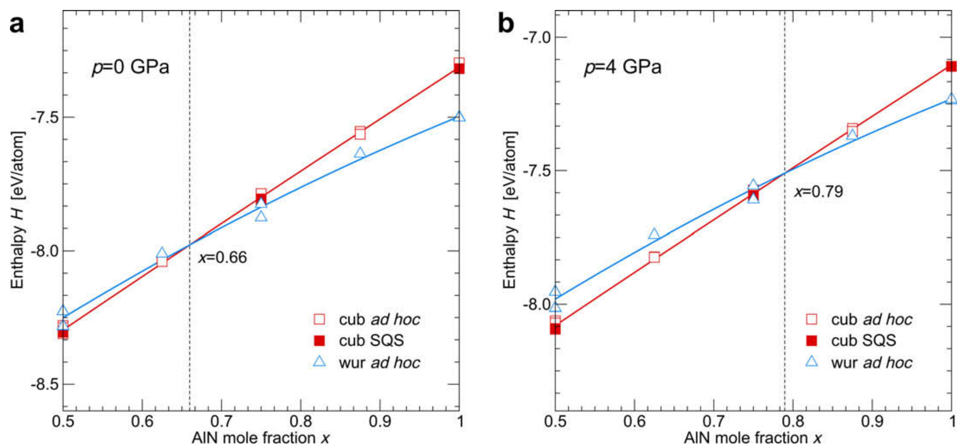

Figure 3. Enthalpies of the cubic (red squares) and wurtzite (blue triangles) phases of the $Cr_{1-x}Al_xN$ system (a) in equilibrium and (b) under a load of 4 GPa hydrostatic pressure. Full and open symbols represent the SQS and the ad hoc configurations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this paper.)

system, x_{max} increases with increasing compressive pressure. The dependence is steeper for $Cr_{1-x}Al_xN$ than in the case of $Ti_{1-x}Al_xN$, especially in the range of tensile hydrostatic pressures. Consequently, for tensile stresses in the coatings x_{max} drops rapidly below 0.5.

In conclusion, we have studied the effect of tensile and compressive hydrostatic pressure on the solubility limit x_{max} of AlN in the cubic B1 phase of $Ti_{1-x}Al_xN$ and $Cr_{1-x}Al_xN$ systems. It follows that x_{max} depends strongly on the applied pressure; compression of about 4 GPa increases x_{max} by ~0.1 AlN mole fraction for both systems. Although the studied stress state differs from the biaxial stress obtained experimentally in thin films during deposition, we can conclude that the built-in stresses influence the phase stabilities of these systems significantly.

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Publication III

Pressure-dependent stability of cubic and wurtzite phases within the TiN-–AlN and CrN—AlN systems

D. Holec, F. Rovere, P.H. Mayrhofer, and P.B. Barna

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Pressure-dependent stability of cubic and wurtzite phases within the TiN–AlN and CrN–AlN systems

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Ab initio calculations are employed to demonstrate a strong pressure dependence of the maximum AlN mole fraction preserving the cubic phase in $Ti_{1-x}Al_xN$ and $Cr_{1-x}Al_xN$ hard coatings before transforming to the wurtzite structure. Under a compression of 4 GPa an increase by ~0.1 AlN mole fraction is obtained for both systems. Consequently, stress-related effects cannot be neglected in the discussion of the wide spread of experimentally obtained stability ranges.

Keywords: Ab initio electron theory; Metastable phases; Coatings; Residual stresses; Nitrides

$Ti_{1-x}Al_xN$ and $Cr_{1-x}Al_xN$ hard coatings are widely used in industrial applications due to their superior mechanical and chemical properties such as high hardness, good wear resistance, and high oxidation and corrosion resistance [1–5]. These excellent properties, which in general improve with increasing Al content, are, however, obtained only for the metastable cubic structure (B1, NaCl-like). Al-rich phases prefer the hexagonal (wurtzite B4, ZnS-like) structure which possesses significantly poorer mechanical properties [1,6]. Therefore knowledge of maximum AlN mole fraction in the cubic phase is of basic interest.

There are several ways to calculate the maximum solubility of one phase in another. Makino [7] used a semi-empirical band parameter method to predict the maximum mole fraction of AlN while maintaining the cubic structure as 0.65 in the $Ti_{1-x}Al_xN$ system and 0.77 in the $Cr_{1-x}Al_xN$ system. Zhang and Veprek [6] used a combined ab initio and thermodynamic approach to obtain the maximum solubility of AlN in $Ti_{1-x}Al_xN$ to be 0.68. Mayrhofer et al. [8,9] employed a fully ab initio approach to calculate energies of formation, E_f, of the cubic B1 and wurtzite B4 phases for several discrete compositions. Assuming that E_f depends smoothly on the composition x, the intersection point of E_f^{cubic} and E_f^{wurtzite} determines the solubility limit. Mayrhofer et al. obtained values of ~0.69 for $Ti_{1-x}Al_xN$ [8] and ~0.66 for $Cr_{1-x}Al_xN$ [9].

The experimental observations of the maximum AlN fraction in the cubic phase in $Cr_{1-x}Al_xN$ and $Ti_{1-x}Al_xN$ are not conclusive. The reports for $Ti_{1-x}Al_xN$ span a wide range from 0.4 [10] to 0.68 [11], 0.8 [12] and 0.91 [13]. Similarly, the solubility limit for $Cr_{1-x}Al_xN$ varies in the range 0.6–0.8 [4,14].

It should be noted that the above mentioned ab initio data correspond to 0 K temperatures. The E_f curves (in particular the wurtzite one) are quite flat around the maximum solubility limit, $x_{\max}$. Thus a change of the order 10^1 meV atom^{-1} between E_f^{cubic} and E_f^{wurtzite} corresponds to a shift of a few per cent in $x_{\max}$ [8,9]. Contributions from the vibrational energy at finite temperatures as well as imperfections of the crystal lattices, which differ from material to material and from phase to phase, may account for the discrepancies between experiments and theory. Mayrhofer et al. [8,9] discussed the effect of a partial clustering and showed that, depending on the actual configuration on the atomic level, the cubic–wurtzite E_f intersection can move by as much as 0.1 AlN mole fraction.

Here we discuss an influence of another important parameter, the pressure p. It has been experimentally shown that stresses in thin films strongly depend on the process conditions, in particular, where the energy of the impinging particles is crucial (e.g. by applying higher bias potential). With increasing bias potential

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(which is generally connected with increasing compressive stresses [15–20]), higher maximum mole fractions of AlN in the cubic phase of $Ti_{1-x}Al_xN$ based materials are obtained [21]. The stresses in the experiments are usually biaxial in character. In this study, however, we consider the hydrostatic pressure for which additional information such as the individual elastic constants, etc., is not needed. It should be noted that positive values of pressure correspond to compression.

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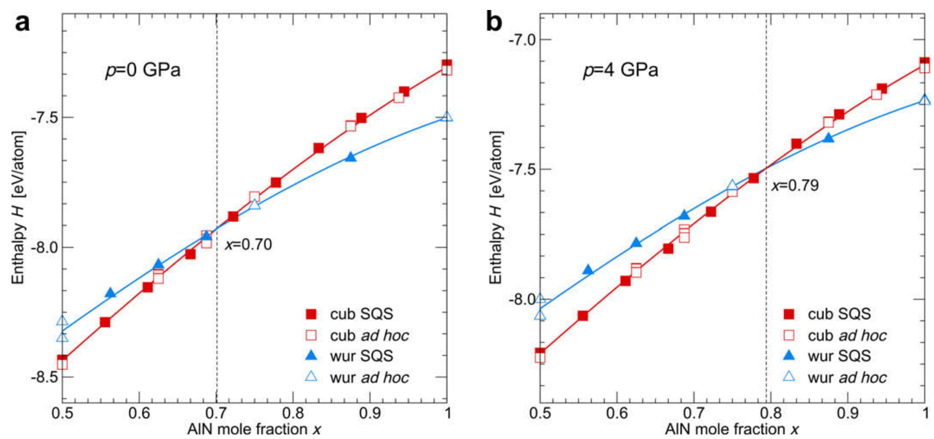

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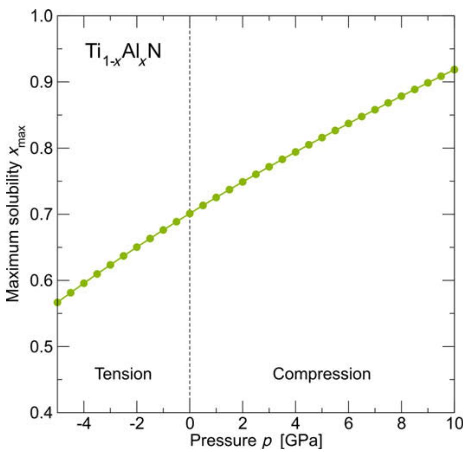

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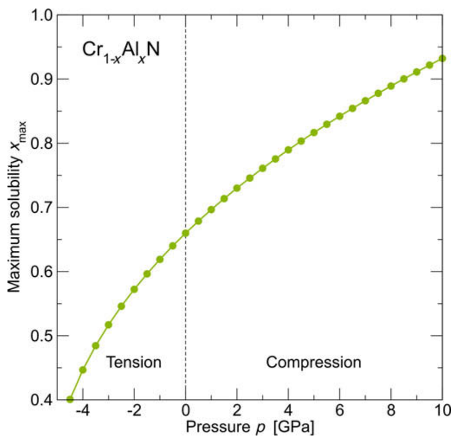

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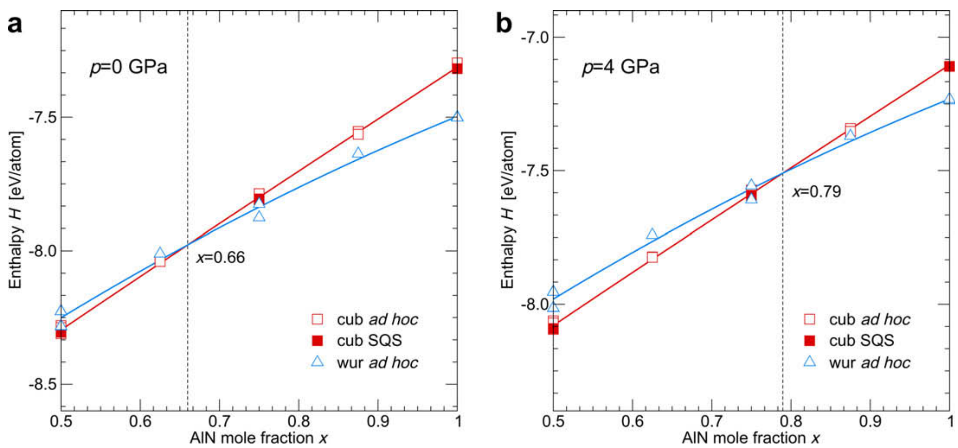

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Publication IV

Pressure-dependent stability of cubic and wurtzite phases within the TiN-–AlN and CrN—AlN systems

D. Holec, F. Rovere, P.H. Mayrhofer, and P.B. Barna

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Ab initio calculations are employed to demonstrate a strong pressure dependence of the maximum AlN mole fraction preserving the cubic phase in $Ti_{1-x}Al_xN$ and $Cr_{1-x}Al_xN$ hard coatings before transforming to the wurtzite structure. Under a compression of 4 GPa an increase by ∼0.1 AlN mole fraction is obtained for both systems. Consequently, stress-related effects cannot be neglected in the discussion of the wide spread of experimentally obtained stability ranges.

Keywords: Ab initio electron theory; Metastable phases; Coatings; Residual stresses; Nitrides

$Ti_{1-x}Al_xN$ and $Cr_{1-x}Al_xN$ hard coatings are widely used in industrial applications due to their superior mechanical and chemical properties such as high hardness, good wear resistance, and high oxidation and corrosion resistance [1–5]. These excellent properties, which in general improve with increasing Al content, are, however, obtained only for the metastable cubic structure (B1, NaCl-like). Al-rich phases prefer the hexagonal (wurtzite B4, ZnS-like) structure which possesses significantly poorer mechanical properties [1,6]. Therefore knowledge of maximum AlN mole fraction in the cubic phase is of basic interest.

There are several ways to calculate the maximum solubility of one phase in another. Makino [7] used a semi-empirical band parameter method to predict the maximum mole fraction of AlN while maintaining the cubic structure as 0.65 in the $Ti_{1-x}Al_xN$ system and 0.77 in the $Cr_{1-x}Al_xN$ system. Zhang and Veprek [6] used a combined ab initio and thermodynamic approach to obtain the maximum solubility of AlN in $Ti_{1-x}Al_xN$ to be 0.68. Mayrhofer et al. [8,9] employed a fully ab initio approach to calculate energies of formation, E_f, of the cubic B1 and wurtzite B4 phases for several discrete compositions. Assuming that E_f depends smoothly on the composition x, the intersection point of E_f^{cubic} and E_f^{wurtzite} determines the solubility limit. Mayrhofer et al. obtained values of ∼0.69 for $Ti_{1-x}Al_xN$ [8] and ∼0.66 for $Cr_{1-x}Al_xN$ [9].

The experimental observations of the maximum AlN fraction in the cubic phase in $Cr_{1-x}Al_xN$ and $Ti_{1-x}Al_xN$ are not conclusive. The reports for $Ti_{1-x}Al_xN$ span a wide range from 0.4 [10] to 0.68 [11], 0.8 [12] and 0.91 [13]. Similarly, the solubility limit for $Cr_{1-x}Al_xN$ varies in the range 0.6–0.8 [4,14].

It should be noted that the above mentioned ab initio data correspond to 0 K temperatures. The E_f curves (in particular the wurtzite one) are quite flat around the maximum solubility limit, $x_{\max}$. Thus a change of the order 10^1 meV atom^{-1} between E_f^{cubic} and E_f^{wurtzite} corresponds to a shift of a few per cent in $x_{\max}$ [8,9]. Contributions from the vibrational energy at finite temperatures as well as imperfections of the crystal lattices, which differ from material to material and from phase to phase, may account for the discrepancies between experiments and theory. Mayrhofer et al. [8,9] discussed the effect of a partial clustering and showed that, depending on the actual configuration on the atomic level, the cubic–wurtzite E_f intersection can move by as much as 0.1 AlN mole fraction.

Here we discuss an influence of another important parameter, the pressure p. It has been experimentally shown that stresses in thin films strongly depend on the process conditions, in particular, where the energy of the impinging particles is crucial (e.g. by applying higher bias potential). With increasing bias potential

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(which is generally connected with increasing compressive stresses [15–20]), higher maximum mole fractions of AlN in the cubic phase of $Ti_{1-x}Al_xN$ based materials are obtained [21]. The stresses in the experiments are usually biaxial in character. In this study, however, we consider the hydrostatic pressure for which additional information such as the individual elastic constants, etc., is not needed. It should be noted that positive values of pressure correspond to compression.

The internal energy, U, is the correct thermodynamic potential at zero pressure and zero temperature. The formation energy per atom, E_f, is defined (e.g. for the $Ti_{1-x}Al_xN$ system) as follows:

$$E_f^{\xi} = E^{\xi}_{Ti_{1-x}Al_xN} - \frac{1}{2}((1-x)E_{Ti} + xE_{Al} + E_N). \tag{1}$$

Here ξ corresponds to either cubic or wurtzite phase, and E_X is the single element total energy corresponding to the internal energy of pure X (crystalline in the case of Al and Ti, or gas molecules in the case of N_2). Consequently, the maximum solubility limit, which is given as $E_f^{\mathrm{cubic}}(x_{\max}) = E_f^{\mathrm{wurtzite}}(x_{\max})$, is equivalent to $U^{\mathrm{cubic}}(x_{\max}) = U^{\mathrm{wurtzite}}(x_{\max})$. When the pressure is considered (still at zero temperature), the enthalpy, $H = U + pV$, must be taken as the correct thermodynamic potential instead of the internal energy U. The Murnaghan equation of state [22]:

$$E(V) = E_0 + \frac{B_0 V}{B_0'}\left(\frac{(V_0/V)^{B_0'}}{B_0'-1} + 1\right) - \frac{B_0 V_0}{B_0'-1} \tag{2}$$

provides an expression for the equilibrium volume as a function of pressure:

$$p = \frac{\partial E}{\partial V} \Rightarrow V(p) = \frac{V_0}{\sqrt[B_0']{\frac{B_0'}{B_0}p + 1}}. \tag{3}$$

Inserting Eq. (3) into Eq. (2) yields:

$$E(p) = E_0 + \frac{B_0 V_0}{B_0'-1}\left(\frac{\frac{p}{B_0}+1}{\sqrt[B_0']{\frac{B_0'}{B_0}p+1}} - 1\right). \tag{4}$$

Combining these results together, an analytical expression for the enthalpy as a function of pressure is obtained:

$$H(p) = E_0 - \frac{B_0 V_0}{B_0'-1} + \frac{pV_0}{\sqrt[B_0']{\frac{B_0'}{B_0}p+1}}\left(\frac{\frac{B_0}{p}+1}{B_0'-1} + 1\right). \tag{5}$$

Since this expression contains the same parameters as the Murnaghan equation of state (Eq. (2)), its quality can be controlled/improved by, for example, calculating additional $E(V)$ data points.

We used the Vienna Ab Initio Simulation Package [23,24] together with the PAW pseudopotentials [25]. The calculation parameters were sufficient to guarantee a precision of several meV atom^{-1} (for details see Refs. [8,9,26]). While the $Ti_{1-x}Al_xN$ system is treated as non-magnetic, the $Cr_{1-x}Al_xN$ data corresponds to the antiferromagnetic (AFM) arrangement [9]. Although CrN crystallizes in the paramagnetic (PM) structure above the Néel temperature (273–283 K) [27], Alling et al. [28] showed that there is no significant difference in energy of formation and other properties between the PM and the AFM state which is the preferred one under the Néel temperature. If the magnetic moments are not considered, the results vary widely from values obtained by experiment [9].

The datapoints in Figure 1a correspond to a collection of several specially designed cells containing 32 and 64 atoms [8]. In addition, new data points for both cubic and wurtzite phase were calculated using special quasi-random structures (SQS) [29]. The third-order polynomials were fitted to the data at a given pressure in order to estimate the crossover of the cubic and wurtzite phase enthalpies. By adding 4 GPa of compressive pressure to the $Ti_{1-x}Al_xN$ system, the crossover between the cubic and hexagonal phases (i.e. the maximum solubility of AlN, $x_{\max}$, in the cubic phase) increases from 0.70 to 0.78 – compare Figure 1a and b, respectively. Consequently, the cubic phase stability increases by ~11%. It is worth noting that compressive stresses around 4 GPa are commonly present in deposited thin

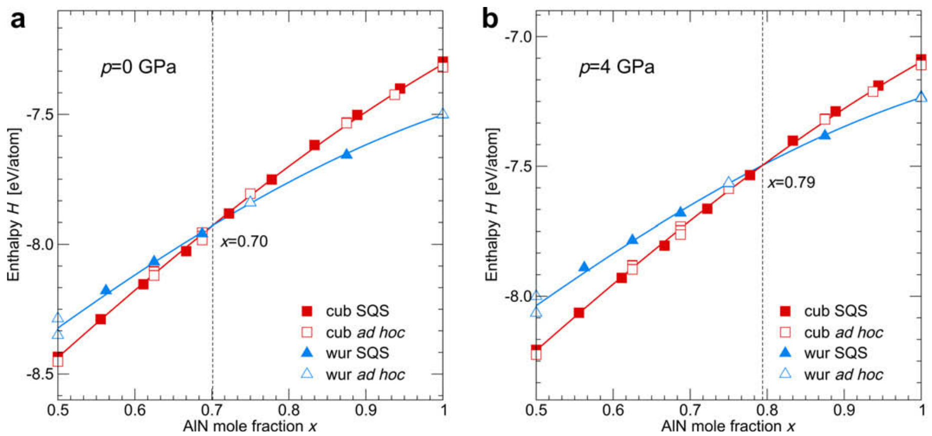

Figure 1. Enthalpies of the cubic (red squares) and wurtzite (blue triangles) phases of the $Ti_{1-x}Al_xN$ system (a) in equilibrium and (b) under a load of 4 GPa hydrostatic pressure. Full and open symbols represent the SQS and the ad hoc configurations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this paper.)

films [20] and thus this model consideration is based on a realistic situation. The dependence of x_{max} in $Ti_{1-x}Al_xN$ as a function of pressure is shown in Figure 2. The trend for increasing the solubility limit with applying larger compressive hydrostatic pressures is predictable since AlN transforms from the wurtzite to the cubic phase at ~16 GPa (a value based on both experiments and calculations) [30,31]. Since the cubic phase has a smaller volume per atom than the wurtzite phase [6], TiN maintains its cubic structure also under compression. Consequently, one would expect that above such pressure the cubic phase is stable for the whole compositional range: indeed, for ~14 GPa the enthalpies of the cubic and wurtzite phases crosses at $x_{max} = 1$, which is in agreement with the value for AlN [30,31].

Subsequently, we applied the same methodology to the $Cr_{1-x}Al_xN$ system. This system is more difficult when compared with $Ti_{1-x}Al_xN$ since the proper magnetic state of the Cr atoms must be taken into account. The difficulties with describing magnetic random alloys are responsible for the absolute values of the ab initio calculated x_{max} to be less accurate than in the case of $Ti_{1-x}Al_xN$.

The data of Mayrhofer et al. [9] used for the phase-stability calculation is based on ad hoc cells. To strengthen our analysis we the added SQS-based data-points of Rovere et al. [26]. In agreement with the $Ti_{1-x}Al_xN$ system, as for the $Cr_{1-x}Al_xN$ system the ad hoc data points fall into the vicinity of the SQS cells (see Figure 3a). The addition of compression increases the maximum AlN solubility of the cubic phase also in the $Cr_{1-x}Al_xN$ system. Under a compressive pressure of 4 GPa, x_{max} is around 0.79 (see Figure 3b) which is ~20% higher than x_{max} of ~0.66 for 0 GPa (see Figure 3a). The calculated maximum solubility of AlN in $Cr_{1-x}Al_xN$ as a function of pressure p is shown in Figure 4. As for the $Ti_{1-x}Al_xN$

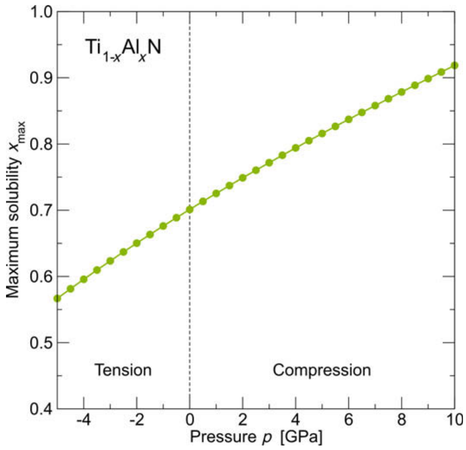

Figure 2. Calculated maximum mole fraction of AlN in the cubic B1 phase of the $Ti_{1-x}Al_xN$ system.

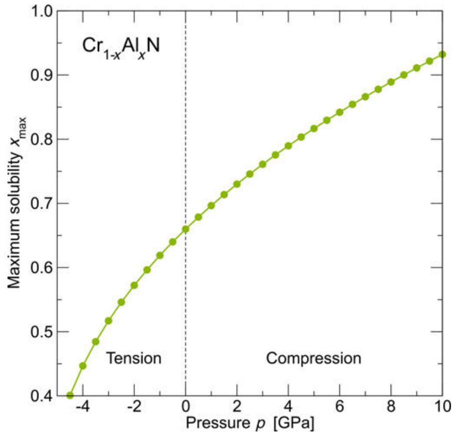

Figure 4. Calculated maximum mole fraction of AlN in the cubic B1 phase of the $Cr_{1-x}Al_xN$ system.

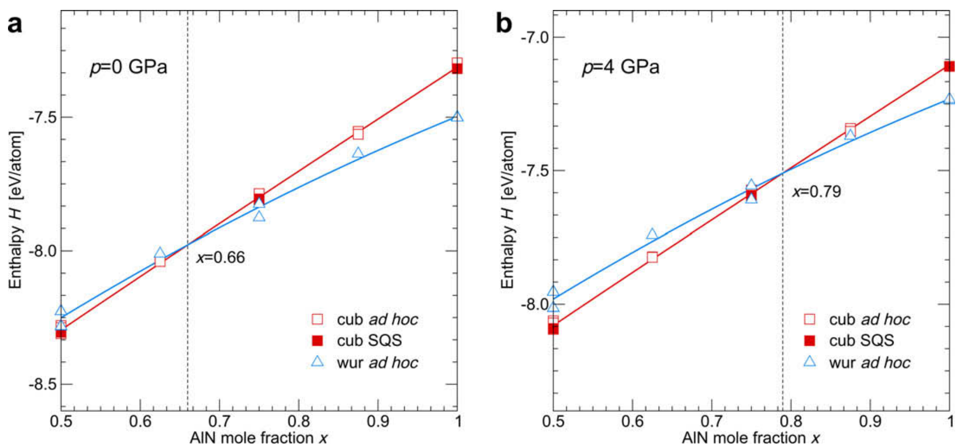

Figure 3. Enthalpies of the cubic (red squares) and wurtzite (blue triangles) phases of the $Cr_{1-x}Al_xN$ system (a) in equilibrium and (b) under a load of 4 GPa hydrostatic pressure. Full and open symbols represent the SQS and the ad hoc configurations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this paper.)

system, x_{max} increases with increasing compressive pressure. The dependence is steeper for $Cr_{1-x}Al_xN$ than in the case of $Ti_{1-x}Al_xN$, especially in the range of tensile hydrostatic pressures. Consequently, for tensile stresses in the coatings x_{max} drops rapidly below 0.5.

In conclusion, we have studied the effect of tensile and compressive hydrostatic pressure on the solubility limit x_{max} of AlN in the cubic B1 phase of $Ti_{1-x}Al_xN$ and $Cr_{1-x}Al_xN$ systems. It follows that x_{max} depends strongly on the applied pressure; compression of about 4 GPa increases x_{max} by $\sim$0.1 AlN mole fraction for both systems. Although the studied stress state differs from the biaxial stress obtained experimentally in thin films during deposition, we can conclude that the built-in stresses influence the phase stabilities of these systems significantly.

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Publication V

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(which is generally connected with increasing compressive stresses [15–20]), higher maximum mole fractions of AlN in the cubic phase of $Ti_{1-x}Al_xN$ based materials are obtained [21]. The stresses in the experiments are usually biaxial in character. In this study, however, we consider the hydrostatic pressure for which additional information such as the individual elastic constants, etc., is not needed. It should be noted that positive values of pressure correspond to compression.

The internal energy, U, is the correct thermodynamic potential at zero pressure and zero temperature. The formation energy per atom, E_f, is defined (e.g. for the $Ti_{1-x}Al_xN$ system) as follows:

$$E_f^{\xi} = E^{\xi}_{Ti_{1-x}Al_xN} - \frac{1}{2}((1-x)E_{Ti} + xE_{Al} + E_N). \tag{1}$$

Here ξ corresponds to either cubic or wurtzite phase, and E_X is the single element total energy corresponding to the internal energy of pure X (crystalline in the case of Al and Ti, or gas molecules in the case of N_2). Consequently, the maximum solubility limit, which is given as $E_f^{\mathrm{cubic}}(x_{\max}) = E_f^{\mathrm{wurtzite}}(x_{\max})$, is equivalent to $U^{\mathrm{cubic}}(x_{\max}) = U^{\mathrm{wurtzite}}(x_{\max})$. When the pressure is considered (still at zero temperature), the enthalpy, $H = U + pV$, must be taken as the correct thermodynamic potential instead of the internal energy U. The Murnaghan equation of state [22]:

$$E(V) = E_0 + \frac{B_0V}{B_0'}\left(\frac{(V_0/V)^{B_0'}}{B_0'-1} + 1\right) - \frac{B_0V_0}{B_0'-1} \tag{2}$$

provides an expression for the equilibrium volume as a function of pressure:

$$p = \frac{\partial E}{\partial V} \Rightarrow V(p) = \frac{V_0}{\sqrt[B_0']{\frac{B_0'}{B_0}p + 1}}. \tag{3}$$

Inserting Eq. (3) into Eq. (2) yields:

$$E(p) = E_0 + \frac{B_0V_0}{B_0'-1}\left(\frac{\frac{p}{B_0} + 1}{\sqrt[B_0']{\frac{B_0'}{B_0}p + 1}} - 1\right). \tag{4}$$

Combining these results together, an analytical expression for the enthalpy as a function of pressure is obtained:

$$H(p) = E_0 - \frac{B_0V_0}{B_0'-1} + \frac{pV_0}{\sqrt[B_0']{\frac{B_0'}{B_0}p + 1}}\left(\frac{\frac{B_0}{p} + 1}{B_0'-1} + 1\right). \tag{5}$$

Since this expression contains the same parameters as the Murnaghan equation of state (Eq. (2)), its quality can be controlled/improved by, for example, calculating additional $E(V)$ data points.

We used the Vienna Ab Initio Simulation Package [23,24] together with the PAW pseudopotentials [25]. The calculation parameters were sufficient to guarantee a precision of several meV atom^{-1} (for details see Refs. [8,9,26]). While the $Ti_{1-x}Al_xN$ system is treated as non-magnetic, the $Cr_{1-x}Al_xN$ data corresponds to the antiferromagnetic (AFM) arrangement [9]. Although CrN crystallizes in the paramagnetic (PM) structure above the Néel temperature (273–283 K) [27], Alling et al. [28] showed that there is no significant difference in energy of formation and other properties between the PM and the AFM state which is the preferred one under the Néel temperature. If the magnetic moments are not considered, the results vary widely from values obtained by experiment [9].

The datapoints in Figure 1a correspond to a collection of several specially designed cells containing 32 and 64 atoms [8]. In addition, new data points for both cubic and wurtzite phase were calculated using special quasi-random structures (SQS) [29]. The third-order polynomials were fitted to the data at a given pressure in order to estimate the crossover of the cubic and wurtzite phase enthalpies. By adding 4 GPa of compressive pressure to the $Ti_{1-x}Al_xN$ system, the crossover between the cubic and hexagonal phases (i.e. the maximum solubility of AlN, $x_{\max}$, in the cubic phase) increases from 0.70 to 0.78 – compare Figure 1a and b, respectively. Consequently, the cubic phase stability increases by ~11%. It is worth noting that compressive stresses around 4 GPa are commonly present in deposited thin

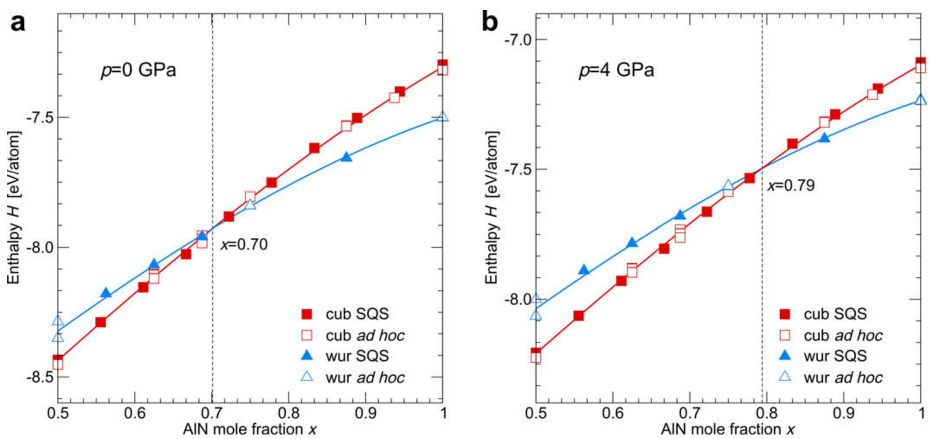

Figure 1. Enthalpies of the cubic (red squares) and wurtzite (blue triangles) phases of the $Ti_{1-x}Al_xN$ system (a) in equilibrium and (b) under a load of 4 GPa hydrostatic pressure. Full and open symbols represent the SQS and the ad hoc configurations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this paper.)

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films [20] and thus this model consideration is based on a realistic situation. The dependence of x_{max} in $Ti_{1-x}Al_xN$ as a function of pressure is shown in Figure 2. The trend for increasing the solubility limit with applying larger compressive hydrostatic pressures is predictable since AlN transforms from the wurtzite to the cubic phase at ~16 GPa (a value based on both experiments and calculations) [30,31]. Since the cubic phase has a smaller volume per atom than the wurtzite phase [6], TiN maintains its cubic structure also under compression. Consequently, one would expect that above such pressure the cubic phase is stable for the whole compositional range: indeed, for ~14 GPa the enthalpies of the cubic and wurtzite phases crosses at $x_{max} = 1$, which is in agreement with the value for AlN [30,31].

Subsequently, we applied the same methodology to the $Cr_{1-x}Al_xN$ system. This system is more difficult when compared with $Ti_{1-x}Al_xN$ since the proper magnetic state of the Cr atoms must be taken into account. The difficulties with describing magnetic random alloys are responsible for the absolute values of the ab initio calculated x_{max} to be less accurate than in the case of $Ti_{1-x}Al_xN$.

The data of Mayrhofer et al. [9] used for the phase-stability calculation is based on ad hoc cells. To strengthen our analysis we the added SQS-based data-points of Rovere et al. [26]. In agreement with the $Ti_{1-x}Al_xN$ system, as for the $Cr_{1-x}Al_xN$ system the ad hoc data points fall into the vicinity of the SQS cells (see Figure 3a). The addition of compression increases the maximum AlN solubility of the cubic phase also in the $Cr_{1-x}Al_xN$ system. Under a compressive pressure of 4 GPa, x_{max} is around 0.79 (see Figure 3b) which is ~20% higher than x_{max} of ~0.66 for 0 GPa (see Figure 3a). The calculated maximum solubility of AlN in $Cr_{1-x}Al_xN$ as a function of pressure p is shown in Figure 4. As for the $Ti_{1-x}Al_xN$

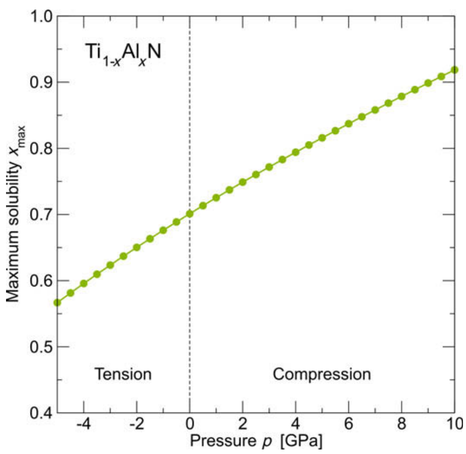

Figure 2. Calculated maximum mole fraction of AlN in the cubic B1 phase of the $Ti_{1-x}Al_xN$ system.

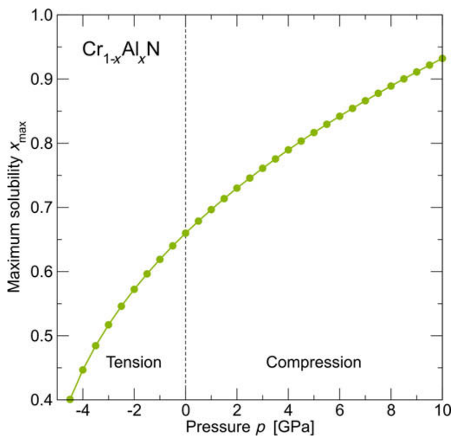

Figure 4. Calculated maximum mole fraction of AlN in the cubic B1 phase of the $Cr_{1-x}Al_xN$ system.

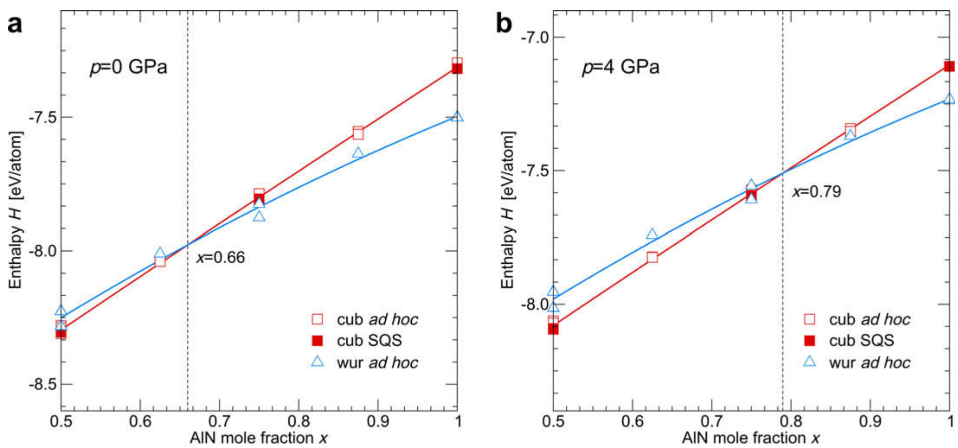

Figure 3. Enthalpies of the cubic (red squares) and wurtzite (blue triangles) phases of the $Cr_{1-x}Al_xN$ system (a) in equilibrium and (b) under a load of 4 GPa hydrostatic pressure. Full and open symbols represent the SQS and the ad hoc configurations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this paper.)

system, x_{max} increases with increasing compressive pressure. The dependence is steeper for $Cr_{1-x}Al_xN$ than in the case of $Ti_{1-x}Al_xN$, especially in the range of tensile hydrostatic pressures. Consequently, for tensile stresses in the coatings x_{max} drops rapidly below 0.5.

In conclusion, we have studied the effect of tensile and compressive hydrostatic pressure on the solubility limit x_{max} of AlN in the cubic B1 phase of $Ti_{1-x}Al_xN$ and $Cr_{1-x}Al_xN$ systems. It follows that x_{max} depends strongly on the applied pressure; compression of about 4 GPa increases x_{max} by $\sim$0.1 AlN mole fraction for both systems. Although the studied stress state differs from the biaxial stress obtained experimentally in thin films during deposition, we can conclude that the built-in stresses influence the phase stabilities of these systems significantly.

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Publication VI

Pressure-dependent stability of cubic and wurtzite phases within the TiN-–AlN and CrN—AlN systems

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Pressure-dependent stability of cubic and wurtzite phases within the TiN–AlN and CrN–AlN systems

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Ab initio calculations are employed to demonstrate a strong pressure dependence of the maximum AlN mole fraction preserving the cubic phase in $Ti_{1-x}Al_xN$ and $Cr_{1-x}Al_xN$ hard coatings before transforming to the wurtzite structure. Under a compression of 4 GPa an increase by ~0.1 AlN mole fraction is obtained for both systems. Consequently, stress-related effects cannot be neglected in the discussion of the wide spread of experimentally obtained stability ranges.

Keywords: Ab initio electron theory; Metastable phases; Coatings; Residual stresses; Nitrides

$Ti_{1-x}Al_xN$ and $Cr_{1-x}Al_xN$ hard coatings are widely used in industrial applications due to their superior mechanical and chemical properties such as high hardness, good wear resistance, and high oxidation and corrosion resistance [1–5]. These excellent properties, which in general improve with increasing Al content, are, however, obtained only for the metastable cubic structure (B1, NaCl-like). Al-rich phases prefer the hexagonal (wurtzite B4, ZnS-like) structure which possesses significantly poorer mechanical properties [1,6]. Therefore knowledge of maximum AlN mole fraction in the cubic phase is of basic interest.

There are several ways to calculate the maximum solubility of one phase in another. Makino [7] used a semi-empirical band parameter method to predict the maximum mole fraction of AlN while maintaining the cubic structure as 0.65 in the $Ti_{1-x}Al_xN$ system and 0.77 in the $Cr_{1-x}Al_xN$ system. Zhang and Veprek [6] used a combined ab initio and thermodynamic approach to obtain the maximum solubility of AlN in $Ti_{1-x}Al_xN$ to be 0.68. Mayrhofer et al. [8,9] employed a fully ab initio approach to calculate energies of formation, E_f, of the cubic B1 and wurtzite B4 phases for several discrete compositions. Assuming that E_f depends smoothly on the composition x, the intersection point of E_f^{cubic} and E_f^{wurtzite} determines the solubility limit. Mayrhofer et al. obtained values of ~0.69 for $Ti_{1-x}Al_xN$ [8] and ~0.66 for $Cr_{1-x}Al_xN$ [9].

The experimental observations of the maximum AlN fraction in the cubic phase in $Cr_{1-x}Al_xN$ and $Ti_{1-x}Al_xN$ are not conclusive. The reports for $Ti_{1-x}Al_xN$ span a wide range from 0.4 [10] to 0.68 [11], 0.8 [12] and 0.91 [13]. Similarly, the solubility limit for $Cr_{1-x}Al_xN$ varies in the range 0.6–0.8 [4,14].

It should be noted that the above mentioned ab initio data correspond to 0 K temperatures. The E_f curves (in particular the wurtzite one) are quite flat around the maximum solubility limit, $x_{\max}$. Thus a change of the order 10^1 meV atom^{-1} between E_f^{cubic} and E_f^{wurtzite} corresponds to a shift of a few per cent in $x_{\max}$ [8,9]. Contributions from the vibrational energy at finite temperatures as well as imperfections of the crystal lattices, which differ from material to material and from phase to phase, may account for the discrepancies between experiments and theory. Mayrhofer et al. [8,9] discussed the effect of a partial clustering and showed that, depending on the actual configuration on the atomic level, the cubic–wurtzite E_f intersection can move by as much as 0.1 AlN mole fraction.

Here we discuss an influence of another important parameter, the pressure p. It has been experimentally shown that stresses in thin films strongly depend on the process conditions, in particular, where the energy of the impinging particles is crucial (e.g. by applying higher bias potential). With increasing bias potential

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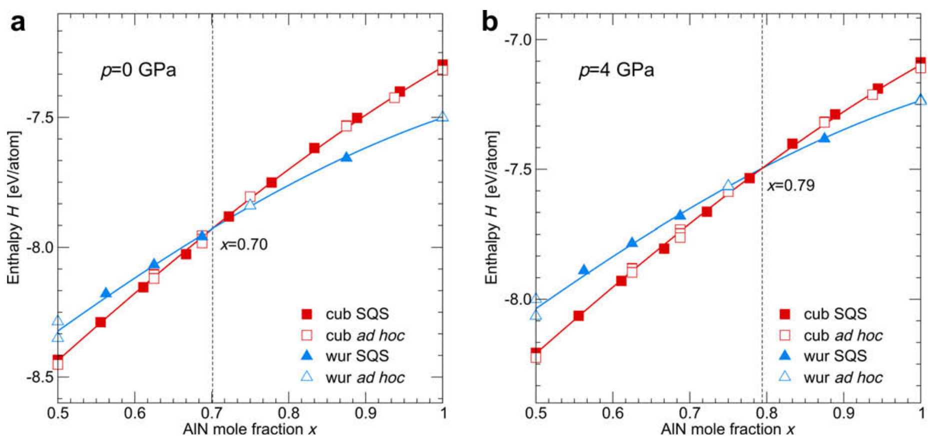

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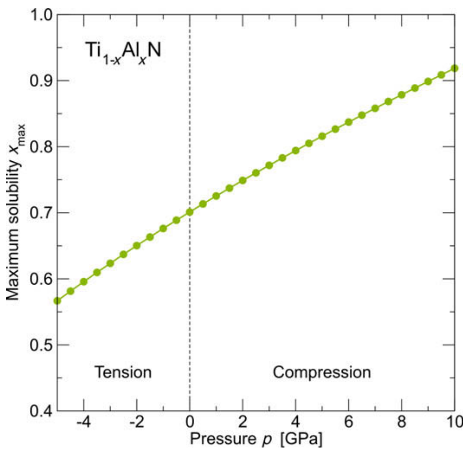

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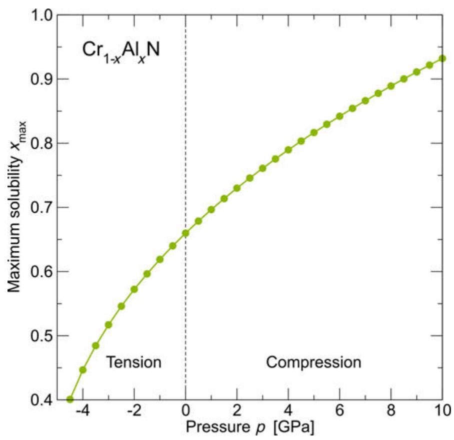

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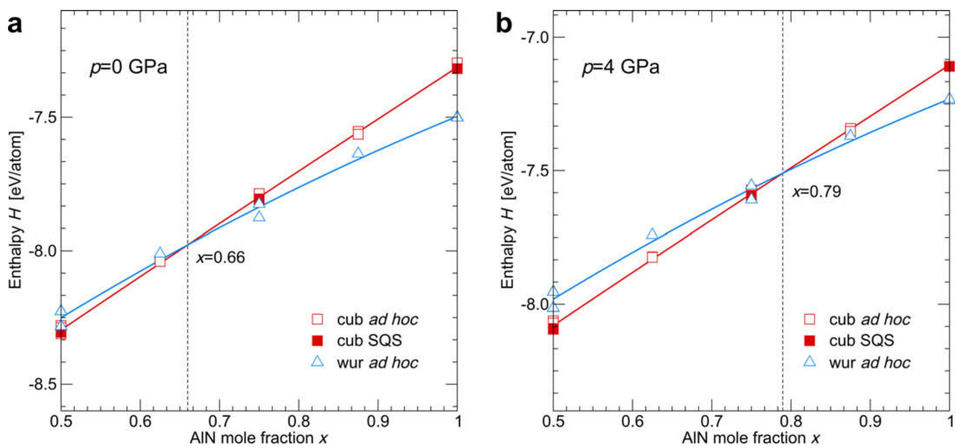

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Publication VII

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Ab initio calculations are employed to demonstrate a strong pressure dependence of the maximum AlN mole fraction preserving the cubic phase in $Ti_{1-x}Al_xN$ and $Cr_{1-x}Al_xN$ hard coatings before transforming to the wurtzite structure. Under a compression of 4 GPa an increase by ~0.1 AlN mole fraction is obtained for both systems. Consequently, stress-related effects cannot be neglected in the discussion of the wide spread of experimentally obtained stability ranges.

Keywords: Ab initio electron theory; Metastable phases; Coatings; Residual stresses; Nitrides

$Ti_{1-x}Al_xN$ and $Cr_{1-x}Al_xN$ hard coatings are widely used in industrial applications due to their superior mechanical and chemical properties such as high hardness, good wear resistance, and high oxidation and corrosion resistance [1–5]. These excellent properties, which in general improve with increasing Al content, are, however, obtained only for the metastable cubic structure (B1, NaCl-like). Al-rich phases prefer the hexagonal (wurtzite B4, ZnS-like) structure which possesses significantly poorer mechanical properties [1,6]. Therefore knowledge of maximum AlN mole fraction in the cubic phase is of basic interest.

There are several ways to calculate the maximum solubility of one phase in another. Makino [7] used a semi-empirical band parameter method to predict the maximum mole fraction of AlN while maintaining the cubic structure as 0.65 in the $Ti_{1-x}Al_xN$ system and 0.77 in the $Cr_{1-x}Al_xN$ system. Zhang and Veprek [6] used a combined ab initio and thermodynamic approach to obtain the maximum solubility of AlN in $Ti_{1-x}Al_xN$ to be 0.68. Mayrhofer et al. [8,9] employed a fully ab initio approach to calculate energies of formation, E_f, of the cubic B1 and wurtzite B4 phases for several discrete compositions. Assuming that E_f depends smoothly on the composition x, the intersection point of E_f^{cubic} and E_f^{wurtzite} determines the solubility limit. Mayrhofer et al. obtained values of ~0.69 for $Ti_{1-x}Al_xN$ [8] and ~0.66 for $Cr_{1-x}Al_xN$ [9].

The experimental observations of the maximum AlN fraction in the cubic phase in $Cr_{1-x}Al_xN$ and $Ti_{1-x}Al_xN$ are not conclusive. The reports for $Ti_{1-x}Al_xN$ span a wide range from 0.4 [10] to 0.68 [11], 0.8 [12] and 0.91 [13]. Similarly, the solubility limit for $Cr_{1-x}Al_xN$ varies in the range 0.6–0.8 [4,14].

It should be noted that the above mentioned ab initio data correspond to 0 K temperatures. The E_f curves (in particular the wurtzite one) are quite flat around the maximum solubility limit, $x_{\max}$. Thus a change of the order 10^1 meV atom^{-1} between E_f^{cubic} and E_f^{wurtzite} corresponds to a shift of a few per cent in $x_{\max}$ [8,9]. Contributions from the vibrational energy at finite temperatures as well as imperfections of the crystal lattices, which differ from material to material and from phase to phase, may account for the discrepancies between experiments and theory. Mayrhofer et al. [8,9] discussed the effect of a partial clustering and showed that, depending on the actual configuration on the atomic level, the cubic–wurtzite E_f intersection can move by as much as 0.1 AlN mole fraction.

Here we discuss an influence of another important parameter, the pressure p. It has been experimentally shown that stresses in thin films strongly depend on the process conditions, in particular, where the energy of the impinging particles is crucial (e.g. by applying higher bias potential). With increasing bias potential

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(which is generally connected with increasing compressive stresses [15–20]), higher maximum mole fractions of AlN in the cubic phase of $Ti_{1-x}Al_xN$ based materials are obtained [21]. The stresses in the experiments are usually biaxial in character. In this study, however, we consider the hydrostatic pressure for which additional information such as the individual elastic constants, etc., is not needed. It should be noted that positive values of pressure correspond to compression.

The internal energy, U, is the correct thermodynamic potential at zero pressure and zero temperature. The formation energy per atom, E_f, is defined (e.g. for the $Ti_{1-x}Al_xN$ system) as follows:

$$E_f^{\xi} = E^{\xi}_{Ti_{1-x}Al_xN} - \frac{1}{2}((1-x)E_{Ti} + xE_{Al} + E_N). \tag{1}$$

Here ξ corresponds to either cubic or wurtzite phase, and E_X is the single element total energy corresponding to the internal energy of pure X (crystalline in the case of Al and Ti, or gas molecules in the case of N_2). Consequently, the maximum solubility limit, which is given as $E_f^{\mathrm{cubic}}(x_{\max}) = E_f^{\mathrm{wurtzite}}(x_{\max})$, is equivalent to $U^{\mathrm{cubic}}(x_{\max}) = U^{\mathrm{wurtzite}}(x_{\max})$. When the pressure is considered (still at zero temperature), the enthalpy, $H = U + pV$, must be taken as the correct thermodynamic potential instead of the internal energy U. The Murnaghan equation of state [22]:

$$E(V) = E_0 + \frac{B_0 V}{B_0'}\left(\frac{(V_0/V)^{B_0'}}{B_0'-1} + 1\right) - \frac{B_0 V_0}{B_0'-1} \tag{2}$$

provides an expression for the equilibrium volume as a function of pressure:

$$p = \frac{\partial E}{\partial V} \Rightarrow V(p) = \frac{V_0}{\sqrt[B_0']{\frac{B_0'}{B_0}p + 1}}. \tag{3}$$

Inserting Eq. (3) into Eq. (2) yields:

$$E(p) = E_0 + \frac{B_0 V_0}{B_0'-1}\left(\frac{\frac{p}{B_0} + 1}{\sqrt[B_0']{\frac{B_0'}{B_0}p + 1}} - 1\right). \tag{4}$$

Combining these results together, an analytical expression for the enthalpy as a function of pressure is obtained:

$$H(p) = E_0 - \frac{B_0 V_0}{B_0'-1} + \frac{pV_0}{\sqrt[B_0']{\frac{B_0'}{B_0}p + 1}}\left(\frac{\frac{B_0}{p} + 1}{B_0'-1} + 1\right). \tag{5}$$

Since this expression contains the same parameters as the Murnaghan equation of state (Eq. (2)), its quality can be controlled/improved by, for example, calculating additional $E(V)$ data points.

We used the Vienna Ab Initio Simulation Package [23,24] together with the PAW pseudopotentials [25]. The calculation parameters were sufficient to guarantee a precision of several meV atom^{-1} (for details see Refs. [8,9,26]). While the $Ti_{1-x}Al_xN$ system is treated as non-magnetic, the $Cr_{1-x}Al_xN$ data corresponds to the antiferromagnetic (AFM) arrangement [9]. Although CrN crystallizes in the paramagnetic (PM) structure above the Néel temperature (273–283 K) [27], Alling et al. [28] showed that there is no significant difference in energy of formation and other properties between the PM and the AFM state which is the preferred one under the Néel temperature. If the magnetic moments are not considered, the results vary widely from values obtained by experiment [9].

The datapoints in Figure 1a correspond to a collection of several specially designed cells containing 32 and 64 atoms [8]. In addition, new data points for both cubic and wurtzite phase were calculated using special quasi-random structures (SQS) [29]. The third-order polynomials were fitted to the data at a given pressure in order to estimate the crossover of the cubic and wurtzite phase enthalpies. By adding 4 GPa of compressive pressure to the $Ti_{1-x}Al_xN$ system, the crossover between the cubic and hexagonal phases (i.e. the maximum solubility of AlN, $x_{\max}$, in the cubic phase) increases from 0.70 to 0.78 – compare Figure 1a and b, respectively. Consequently, the cubic phase stability increases by ~11%. It is worth noting that compressive stresses around 4 GPa are commonly present in deposited thin

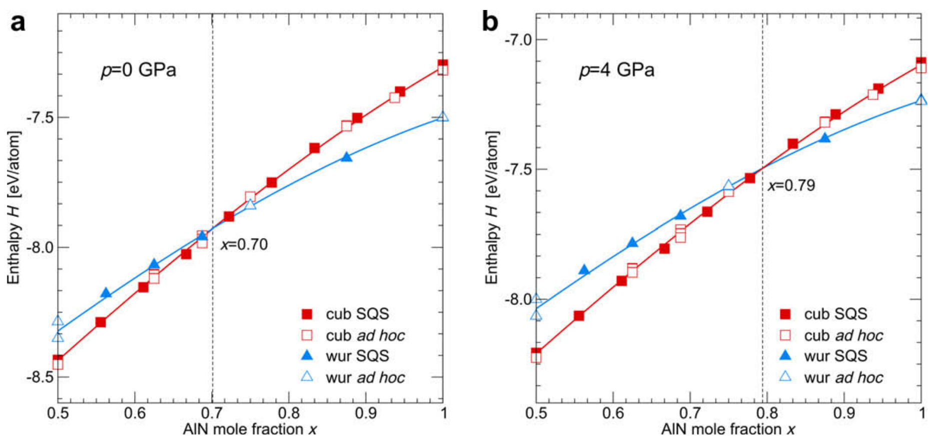

Figure 1. Enthalpies of the cubic (red squares) and wurtzite (blue triangles) phases of the $Ti_{1-x}Al_xN$ system (a) in equilibrium and (b) under a load of 4 GPa hydrostatic pressure. Full and open symbols represent the SQS and the ad hoc configurations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this paper.)

films [20] and thus this model consideration is based on a realistic situation. The dependence of x_{max} in $Ti_{1-x}Al_xN$ as a function of pressure is shown in Figure 2. The trend for increasing the solubility limit with applying larger compressive hydrostatic pressures is predictable since AlN transforms from the wurtzite to the cubic phase at ∼16 GPa (a value based on both experiments and calculations) [30,31]. Since the cubic phase has a smaller volume per atom than the wurtzite phase [6], TiN maintains its cubic structure also under compression. Consequently, one would expect that above such pressure the cubic phase is stable for the whole compositional range: indeed, for ∼14 GPa the enthalpies of the cubic and wurtzite phases crosses at $x_{max} = 1$, which is in agreement with the value for AlN [30,31].

Subsequently, we applied the same methodology to the $Cr_{1-x}Al_xN$ system. This system is more difficult when compared with $Ti_{1-x}Al_xN$ since the proper magnetic state of the Cr atoms must be taken into account. The difficulties with describing magnetic random alloys are responsible for the absolute values of the ab initio calculated x_{max} to be less accurate than in the case of $Ti_{1-x}Al_xN$.

The data of Mayrhofer et al. [9] used for the phase-stability calculation is based on ad hoc cells. To strengthen our analysis we the added SQS-based data-points of Rovere et al. [26]. In agreement with the $Ti_{1-x}Al_xN$ system, as for the $Cr_{1-x}Al_xN$ system the ad hoc data points fall into the vicinity of the SQS cells (see Figure 3a). The addition of compression increases the maximum AlN solubility of the cubic phase also in the $Cr_{1-x}Al_xN$ system. Under a compressive pressure of 4 GPa, x_{max} is around 0.79 (see Figure 3b) which is ∼20% higher than x_{max} of ∼0.66 for 0 GPa (see Figure 3a). The calculated maximum solubility of AlN in $Cr_{1-x}Al_xN$ as a function of pressure p is shown in Figure 4. As for the $Ti_{1-x}Al_xN$

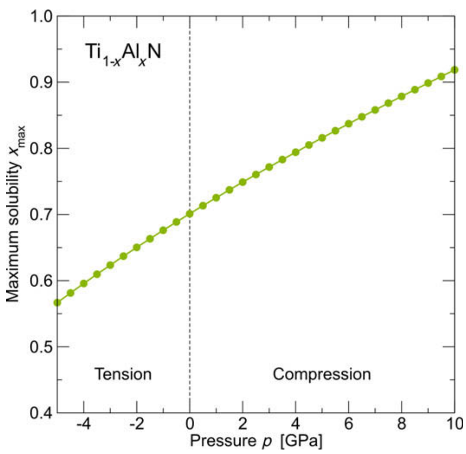

Figure 2. Calculated maximum mole fraction of AlN in the cubic B1 phase of the $Ti_{1-x}Al_xN$ system.

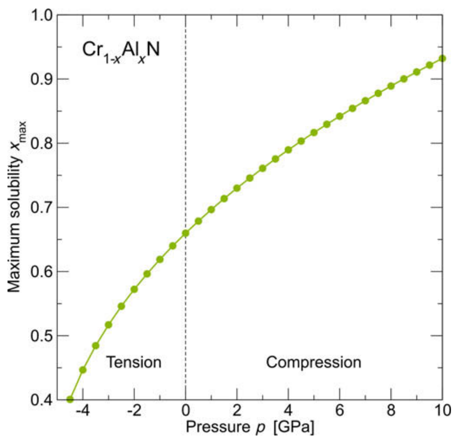

Figure 4. Calculated maximum mole fraction of AlN in the cubic B1 phase of the $Cr_{1-x}Al_xN$ system.

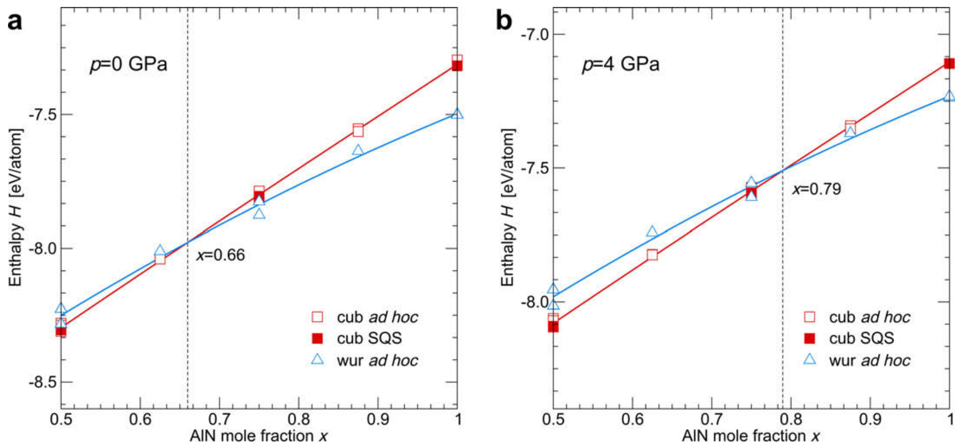

Figure 3. Enthalpies of the cubic (red squares) and wurtzite (blue triangles) phases of the $Cr_{1-x}Al_xN$ system (a) in equilibrium and (b) under a load of 4 GPa hydrostatic pressure. Full and open symbols represent the SQS and the ad hoc configurations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this paper.)

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system, x_{max} increases with increasing compressive pressure. The dependence is steeper for $Cr_{1-x}Al_xN$ than in the case of $Ti_{1-x}Al_xN$, especially in the range of tensile hydrostatic pressures. Consequently, for tensile stresses in the coatings x_{max} drops rapidly below 0.5.

In conclusion, we have studied the effect of tensile and compressive hydrostatic pressure on the solubility limit x_{max} of AlN in the cubic B1 phase of $Ti_{1-x}Al_xN$ and $Cr_{1-x}Al_xN$ systems. It follows that x_{max} depends strongly on the applied pressure; compression of about 4 GPa increases x_{max} by ∼0.1 AlN mole fraction for both systems. Although the studied stress state differs from the biaxial stress obtained experimentally in thin films during deposition, we can conclude that the built-in stresses influence the phase stabilities of these systems significantly.

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Publication VIII

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The internal energy, U, is the correct thermodynamic potential at zero pressure and zero temperature. The formation energy per atom, E_f, is defined (e.g. for the $Ti_{1-x}Al_xN$ system) as follows:

$$E_f^{\xi} = E^{\xi}_{\mathrm{Ti}_{1-x}\mathrm{Al}_x\mathrm{N}} - \frac{1}{2}((1-x)E_{\mathrm{Ti}} + xE_{\mathrm{Al}} + E_{\mathrm{N}}). \tag{1}$$

Here ξ corresponds to either cubic or wurtzite phase, and E_X is the single element total energy corresponding to the internal energy of pure X (crystalline in the case of Al and Ti, or gas molecules in the case of N_2). Consequently, the maximum solubility limit, which is given as $E_f^{\mathrm{cubic}}(x_{\max}) = E_f^{\mathrm{wurtzite}}(x_{\max})$, is equivalent to $U^{\mathrm{cubic}}(x_{\max}) = U^{\mathrm{wurtzite}}(x_{\max})$. When the pressure is considered (still at zero temperature), the enthalpy, $H = U + pV$, must be taken as the correct thermodynamic potential instead of the internal energy U. The Murnaghan equation of state [22]:

$$E(V) = E_0 + \frac{B_0 V}{B_0'}\left(\frac{(V_0/V)^{B_0'}}{B_0'-1} + 1\right) - \frac{B_0 V_0}{B_0'-1} \tag{2}$$

provides an expression for the equilibrium volume as a function of pressure:

$$p = \frac{\partial E}{\partial V} \Rightarrow V(p) = \frac{V_0}{\sqrt[B_0']{\frac{B_0'}{B_0}p + 1}}. \tag{3}$$

Inserting Eq. (3) into Eq. (2) yields:

$$E(p) = E_0 + \frac{B_0 V_0}{B_0'-1}\left(\frac{\frac{p}{B_0} + 1}{\sqrt[B_0']{\frac{B_0'}{B_0}p + 1}} - 1\right). \tag{4}$$

Combining these results together, an analytical expression for the enthalpy as a function of pressure is obtained:

$$H(p) = E_0 - \frac{B_0 V_0}{B_0'-1} + \frac{pV_0}{\sqrt[B_0']{\frac{B_0'}{B_0}p + 1}}\left(\frac{\frac{B_0}{p} + 1}{B_0'-1} + 1\right). \tag{5}$$

Since this expression contains the same parameters as the Murnaghan equation of state (Eq. (2)), its quality can be controlled/improved by, for example, calculating additional $E(V)$ data points.

We used the Vienna Ab Initio Simulation Package [23,24] together with the PAW pseudopotentials [25]. The calculation parameters were sufficient to guarantee a precision of several meV atom^{-1} (for details see Refs. [8,9,26]). While the $Ti_{1-x}Al_xN$ system is treated as non-magnetic, the $Cr_{1-x}Al_xN$ data corresponds to the anti-ferromagnetic (AFM) arrangement [9]. Although CrN crystallizes in the paramagnetic (PM) structure above the Néel temperature (273–283 K) [27], Alling et al. [28] showed that there is no significant difference in energy of formation and other properties between the PM and the AFM state which is the preferred one under the Néel temperature. If the magnetic moments are not considered, the results vary widely from values obtained by experiment [9].

The datapoints in Figure 1a correspond to a collection of several specially designed cells containing 32 and 64 atoms [8]. In addition, new data points for both cubic and wurtzite phase were calculated using special quasi-random structures (SQS) [29]. The third-order polynomials were fitted to the data at a given pressure in order to estimate the crossover of the cubic and wurtzite phase enthalpies. By adding 4 GPa of compressive pressure to the $Ti_{1-x}Al_xN$ system, the crossover between the cubic and hexagonal phases (i.e. the maximum solubility of AlN, $x_{\max}$, in the cubic phase) increases from 0.70 to 0.78 – compare Figure 1a and b, respectively. Consequently, the cubic phase stability increases by ~11%. It is worth noting that compressive stresses around 4 GPa are commonly present in deposited thin

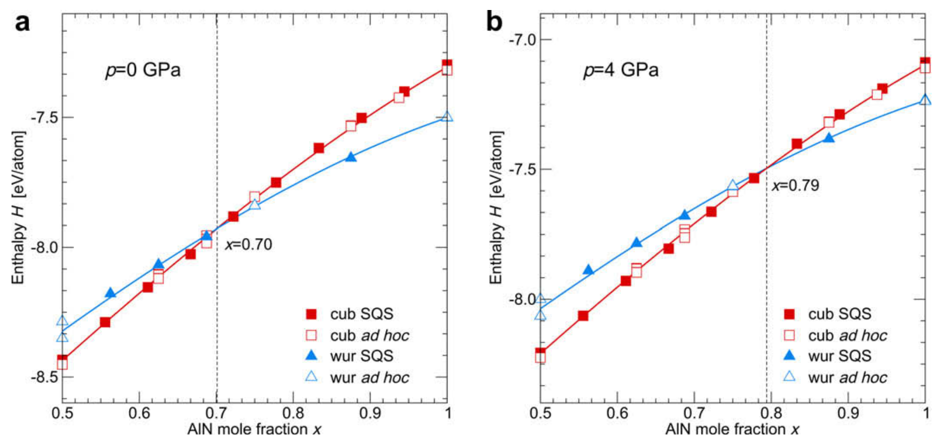

Figure 1. Enthalpies of the cubic (red squares) and wurtzite (blue triangles) phases of the $Ti_{1-x}Al_xN$ system (a) in equilibrium and (b) under a load of 4 GPa hydrostatic pressure. Full and open symbols represent the SQS and the ad hoc configurations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this paper.)

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Subsequently, we applied the same methodology to the $Cr_{1-x}Al_xN$ system. This system is more difficult when compared with $Ti_{1-x}Al_xN$ since the proper magnetic state of the Cr atoms must be taken into account. The difficulties with describing magnetic random alloys are responsible for the absolute values of the ab initio calculated x_{max} to be less accurate than in the case of $Ti_{1-x}Al_xN$.

The data of Mayrhofer et al. [9] used for the phase-stability calculation is based on ad hoc cells. To strengthen our analysis we the added SQS-based data-points of Rovere et al. [26]. In agreement with the $Ti_{1-x}Al_xN$ system, as for the $Cr_{1-x}Al_xN$ system the ad hoc data points fall into the vicinity of the SQS cells (see Figure 3a). The addition of compression increases the maximum AlN solubility of the cubic phase also in the $Cr_{1-x}Al_xN$ system. Under a compressive pressure of 4 GPa, x_{max} is around 0.79 (see Figure 3b) which is ~20% higher than x_{max} of ~0.66 for 0 GPa (see Figure 3a). The calculated maximum solubility of AlN in $Cr_{1-x}Al_xN$ as a function of pressure p is shown in Figure 4. As for the $Ti_{1-x}Al_xN$

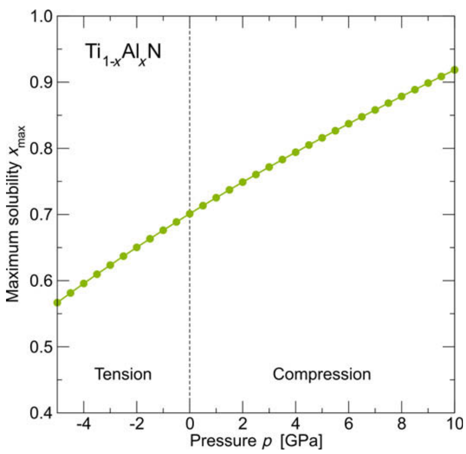

Figure 2. Calculated maximum mole fraction of AlN in the cubic B1 phase of the $Ti_{1-x}Al_xN$ system.

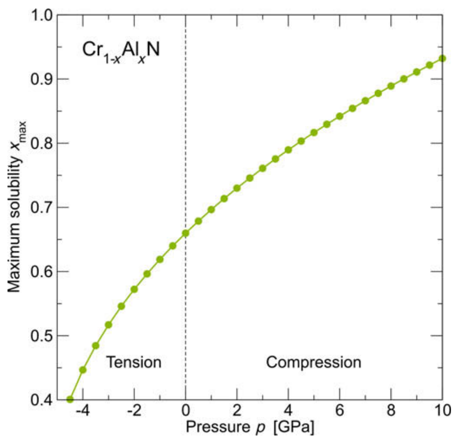

Figure 4. Calculated maximum mole fraction of AlN in the cubic B1 phase of the $Cr_{1-x}Al_xN$ system.

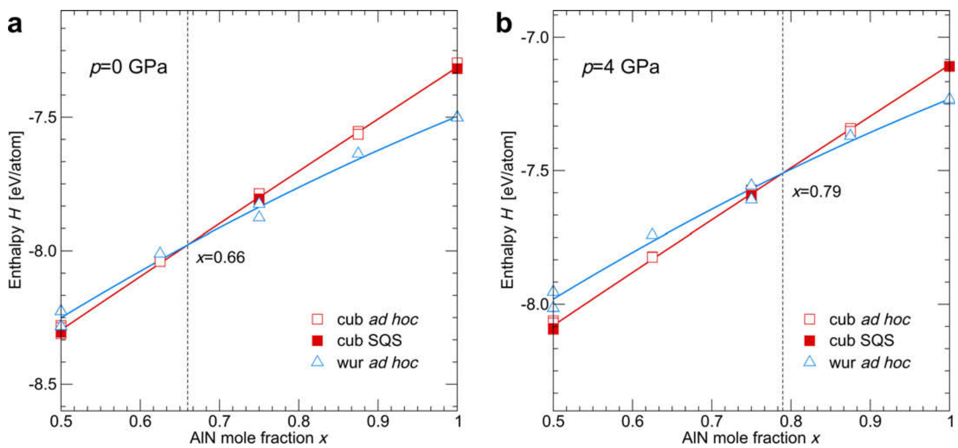

Figure 3. Enthalpies of the cubic (red squares) and wurtzite (blue triangles) phases of the $Cr_{1-x}Al_xN$ system (a) in equilibrium and (b) under a load of 4 GPa hydrostatic pressure. Full and open symbols represent the SQS and the ad hoc configurations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this paper.)

system, x_{max} increases with increasing compressive pressure. The dependence is steeper for $Cr_{1-x}Al_xN$ than in the case of $Ti_{1-x}Al_xN$, especially in the range of tensile hydrostatic pressures. Consequently, for tensile stresses in the coatings x_{max} drops rapidly below 0.5.

In conclusion, we have studied the effect of tensile and compressive hydrostatic pressure on the solubility limit x_{max} of AlN in the cubic B1 phase of $Ti_{1-x}Al_xN$ and $Cr_{1-x}Al_xN$ systems. It follows that x_{max} depends strongly on the applied pressure; compression of about 4 GPa increases x_{max} by ~0.1 AlN mole fraction for both systems. Although the studied stress state differs from the biaxial stress obtained experimentally in thin films during deposition, we can conclude that the built-in stresses influence the phase stabilities of these systems significantly.

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Publication IX

Pressure-dependent stability of cubic and wurtzite phases within the TiN-–AlN and CrN—AlN systems

D. Holec, F. Rovere, P.H. Mayrhofer, and P.B. Barna

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Pressure-dependent stability of cubic and wurtzite phases within the TiN–AlN and CrN–AlN systems

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Ab initio calculations are employed to demonstrate a strong pressure dependence of the maximum AlN mole fraction preserving the cubic phase in $Ti_{1-x}Al_xN$ and $Cr_{1-x}Al_xN$ hard coatings before transforming to the wurtzite structure. Under a compression of 4 GPa an increase by ~0.1 AlN mole fraction is obtained for both systems. Consequently, stress-related effects cannot be neglected in the discussion of the wide spread of experimentally obtained stability ranges.

Keywords: Ab initio electron theory; Metastable phases; Coatings; Residual stresses; Nitrides

$Ti_{1-x}Al_xN$ and $Cr_{1-x}Al_xN$ hard coatings are widely used in industrial applications due to their superior mechanical and chemical properties such as high hardness, good wear resistance, and high oxidation and corrosion resistance [1–5]. These excellent properties, which in general improve with increasing Al content, are, however, obtained only for the metastable cubic structure (B1, NaCl-like). Al-rich phases prefer the hexagonal (wurtzite B4, ZnS-like) structure which possesses significantly poorer mechanical properties [1,6]. Therefore knowledge of maximum AlN mole fraction in the cubic phase is of basic interest.

There are several ways to calculate the maximum solubility of one phase in another. Makino [7] used a semi-empirical band parameter method to predict the maximum mole fraction of AlN while maintaining the cubic structure as 0.65 in the $Ti_{1-x}Al_xN$ system and 0.77 in the $Cr_{1-x}Al_xN$ system. Zhang and Veprek [6] used a combined ab initio and thermodynamic approach to obtain the maximum solubility of AlN in $Ti_{1-x}Al_xN$ to be 0.68. Mayrhofer et al. [8,9] employed a fully ab initio approach to calculate energies of formation, E_f, of the cubic B1 and wurtzite B4 phases for several discrete compositions. Assuming that E_f depends smoothly on the composition x, the intersection point of E_f^{cubic} and E_f^{wurtzite} determines the solubility limit. Mayrhofer et al. obtained values of ~0.69 for $Ti_{1-x}Al_xN$ [8] and ~0.66 for $Cr_{1-x}Al_xN$ [9].

The experimental observations of the maximum AlN fraction in the cubic phase in $Cr_{1-x}Al_xN$ and $Ti_{1-x}Al_xN$ are not conclusive. The reports for $Ti_{1-x}Al_xN$ span a wide range from 0.4 [10] to 0.68 [11], 0.8 [12] and 0.91 [13]. Similarly, the solubility limit for $Cr_{1-x}Al_xN$ varies in the range 0.6–0.8 [4,14].

It should be noted that the above mentioned ab initio data correspond to 0 K temperatures. The E_f curves (in particular the wurtzite one) are quite flat around the maximum solubility limit, $x_{\max}$. Thus a change of the order 10^1 meV atom^{-1} between E_f^{cubic} and E_f^{wurtzite} corresponds to a shift of a few per cent in $x_{\max}$ [8,9]. Contributions from the vibrational energy at finite temperatures as well as imperfections of the crystal lattices, which differ from material to material and from phase to phase, may account for the discrepancies between experiments and theory. Mayrhofer et al. [8,9] discussed the effect of a partial clustering and showed that, depending on the actual configuration on the atomic level, the cubic–wurtzite E_f intersection can move by as much as 0.1 AlN mole fraction.

Here we discuss an influence of another important parameter, the pressure p. It has been experimentally shown that stresses in thin films strongly depend on the process conditions, in particular, where the energy of the impinging particles is crucial (e.g. by applying higher bias potential). With increasing bias potential

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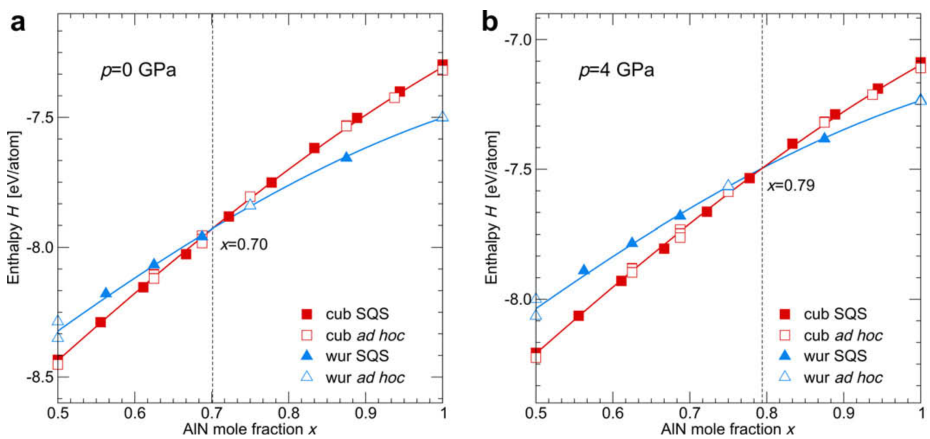

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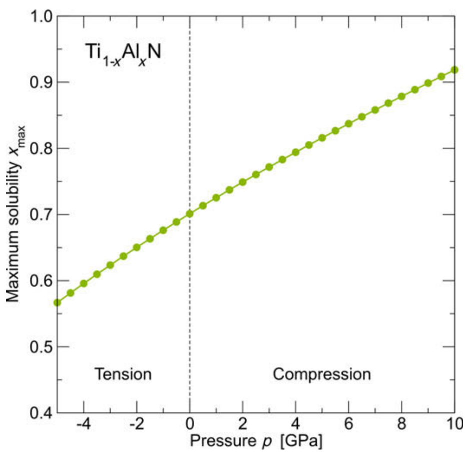

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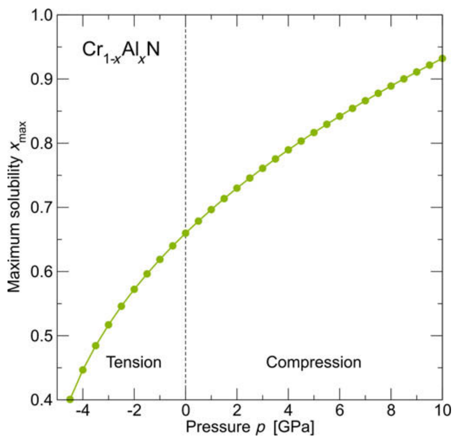

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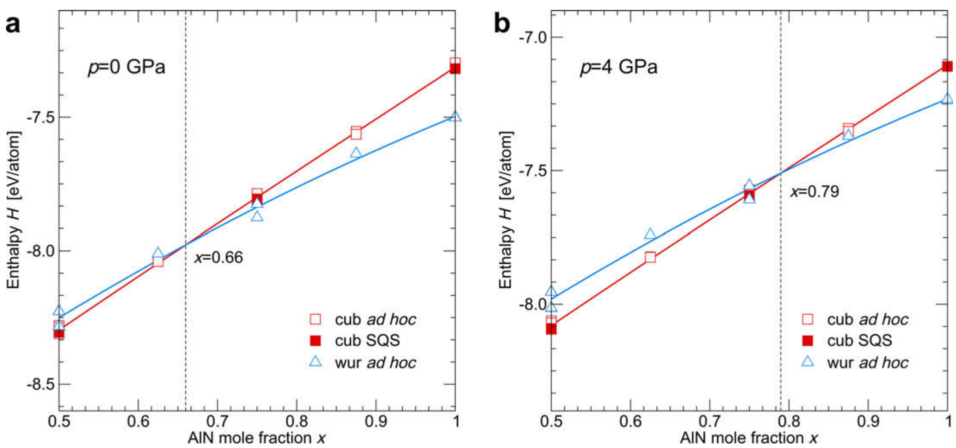

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Publication X

Pressure-dependent stability of cubic and wurtzite phases within the TiN-–AlN and CrN—AlN systems

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Pressure-dependent stability of cubic and wurtzite phases within the TiN–AlN and CrN–AlN systems

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Ab initio calculations are employed to demonstrate a strong pressure dependence of the maximum AlN mole fraction preserving the cubic phase in $Ti_{1-x}Al_xN$ and $Cr_{1-x}Al_xN$ hard coatings before transforming to the wurtzite structure. Under a compression of 4 GPa an increase by ~0.1 AlN mole fraction is obtained for both systems. Consequently, stress-related effects cannot be neglected in the discussion of the wide spread of experimentally obtained stability ranges.

Keywords: Ab initio electron theory; Metastable phases; Coatings; Residual stresses; Nitrides

$Ti_{1-x}Al_xN$ and $Cr_{1-x}Al_xN$ hard coatings are widely used in industrial applications due to their superior mechanical and chemical properties such as high hardness, good wear resistance, and high oxidation and corrosion resistance [1–5]. These excellent properties, which in general improve with increasing Al content, are, however, obtained only for the metastable cubic structure (B1, NaCl-like). Al-rich phases prefer the hexagonal (wurtzite B4, ZnS-like) structure which possesses significantly poorer mechanical properties [1,6]. Therefore knowledge of maximum AlN mole fraction in the cubic phase is of basic interest.

There are several ways to calculate the maximum solubility of one phase in another. Makino [7] used a semi-empirical band parameter method to predict the maximum mole fraction of AlN while maintaining the cubic structure as 0.65 in the $Ti_{1-x}Al_xN$ system and 0.77 in the $Cr_{1-x}Al_xN$ system. Zhang and Veprek [6] used a combined ab initio and thermodynamic approach to obtain the maximum solubility of AlN in $Ti_{1-x}Al_xN$ to be 0.68. Mayrhofer et al. [8,9] employed a fully ab initio approach to calculate energies of formation, E_f, of the cubic B1 and wurtzite B4 phases for several discrete compositions. Assuming that E_f depends smoothly on the composition x, the intersection point of E_f^{cubic} and E_f^{wurtzite} determines the solubility limit. Mayrhofer et al. obtained values of ~0.69 for $Ti_{1-x}Al_xN$ [8] and ~0.66 for $Cr_{1-x}Al_xN$ [9].

The experimental observations of the maximum AlN fraction in the cubic phase in $Cr_{1-x}Al_xN$ and $Ti_{1-x}Al_xN$ are not conclusive. The reports for $Ti_{1-x}Al_xN$ span a wide range from 0.4 [10] to 0.68 [11], 0.8 [12] and 0.91 [13]. Similarly, the solubility limit for $Cr_{1-x}Al_xN$ varies in the range 0.6–0.8 [4,14].

It should be noted that the above mentioned ab initio data correspond to 0 K temperatures. The E_f curves (in particular the wurtzite one) are quite flat around the maximum solubility limit, $x_{\max}$. Thus a change of the order 10^1 meV atom^{-1} between E_f^{cubic} and E_f^{wurtzite} corresponds to a shift of a few per cent in $x_{\max}$ [8,9]. Contributions from the vibrational energy at finite temperatures as well as imperfections of the crystal lattices, which differ from material to material and from phase to phase, may account for the discrepancies between experiments and theory. Mayrhofer et al. [8,9] discussed the effect of a partial clustering and showed that, depending on the actual configuration on the atomic level, the cubic–wurtzite E_f intersection can move by as much as 0.1 AlN mole fraction.

Here we discuss an influence of another important parameter, the pressure p. It has been experimentally shown that stresses in thin films strongly depend on the process conditions, in particular, where the energy of the impinging particles is crucial (e.g. by applying higher bias potential). With increasing bias potential

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(which is generally connected with increasing compressive stresses [15–20]), higher maximum mole fractions of AlN in the cubic phase of $Ti_{1-x}Al_xN$ based materials are obtained [21]. The stresses in the experiments are usually biaxial in character. In this study, however, we consider the hydrostatic pressure for which additional information such as the individual elastic constants, etc., is not needed. It should be noted that positive values of pressure correspond to compression.

The internal energy, U, is the correct thermodynamic potential at zero pressure and zero temperature. The formation energy per atom, E_f, is defined (e.g. for the $Ti_{1-x}Al_xN$ system) as follows:

$$E_f^{\xi} = E^{\xi}_{\mathrm{Ti}_{1-x}\mathrm{Al}_x\mathrm{N}} - \frac{1}{2}((1-x)E_{\mathrm{Ti}} + xE_{\mathrm{Al}} + E_{\mathrm{N}}). \quad (1)$$

Here ξ corresponds to either cubic or wurtzite phase, and E_{X} is the single element total energy corresponding to the internal energy of pure X (crystalline in the case of Al and Ti, or gas molecules in the case of N_2). Consequently, the maximum solubility limit, which is given as $E_f^{\mathrm{cubic}}(x_{\max}) = E_f^{\mathrm{wurtzite}}(x_{\max})$, is equivalent to $U^{\mathrm{cubic}}(x_{\max}) = U^{\mathrm{wurtzite}}(x_{\max})$. When the pressure is considered (still at zero temperature), the enthalpy, $H = U + pV$, must be taken as the correct thermodynamic potential instead of the internal energy U. The Murnaghan equation of state [22]:

$$E(V) = E_0 + \frac{B_0 V}{B_0'}\left(\frac{(V_0/V)^{B_0'}}{B_0'-1} + 1\right) - \frac{B_0 V_0}{B_0'-1} \quad (2)$$

provides an expression for the equilibrium volume as a function of pressure:

$$p = \frac{\partial E}{\partial V} \Rightarrow V(p) = \frac{V_0}{\sqrt[B_0']{\frac{B_0'}{B_0}p + 1}}. \quad (3)$$

Inserting Eq. (3) into Eq. (2) yields:

$$E(p) = E_0 + \frac{B_0 V_0}{B_0'-1}\left(\frac{\frac{p}{B_0}+1}{\sqrt[B_0']{\frac{B_0'}{B_0}p+1}} - 1\right). \quad (4)$$

Combining these results together, an analytical expression for the enthalpy as a function of pressure is obtained:

$$H(p) = E_0 - \frac{B_0 V_0}{B_0'-1} + \frac{pV_0}{\sqrt[B_0']{\frac{B_0'}{B_0}p+1}}\left(\frac{\frac{B_0}{p}+1}{B_0'-1} + 1\right). \quad (5)$$

Since this expression contains the same parameters as the Murnaghan equation of state (Eq. (2)), its quality can be controlled/improved by, for example, calculating additional $E(V)$ data points.

We used the Vienna Ab Initio Simulation Package [23,24] together with the PAW pseudopotentials [25]. The calculation parameters were sufficient to guarantee a precision of several meV atom^{-1} (for details see Refs. [8,9,26]). While the $Ti_{1-x}Al_xN$ system is treated as nonmagnetic, the $Cr_{1-x}Al_xN$ data corresponds to the antiferromagnetic (AFM) arrangement [9]. Although CrN crystallizes in the paramagnetic (PM) structure above the Néel temperature (273–283 K) [27], Alling et al. [28] showed that there is no significant difference in energy of formation and other properties between the PM and the AFM state which is the preferred one under the Néel temperature. If the magnetic moments are not considered, the results vary widely from values obtained by experiment [9].

The datapoints in Figure 1a correspond to a collection of several specially designed cells containing 32 and 64 atoms [8]. In addition, new data points for both cubic and wurtzite phase were calculated using special quasi-random structures (SQS) [29]. The third-order polynomials were fitted to the data at a given pressure in order to estimate the crossover of the cubic and wurtzite phase enthalpies. By adding 4 GPa of compressive pressure to the $Ti_{1-x}Al_xN$ system, the crossover between the cubic and hexagonal phases (i.e. the maximum solubility of AlN, $x_{\max}$, in the cubic phase) increases from 0.70 to 0.78 – compare Figure 1a and b, respectively. Consequently, the cubic phase stability increases by ~11%. It is worth noting that compressive stresses around 4 GPa are commonly present in deposited thin

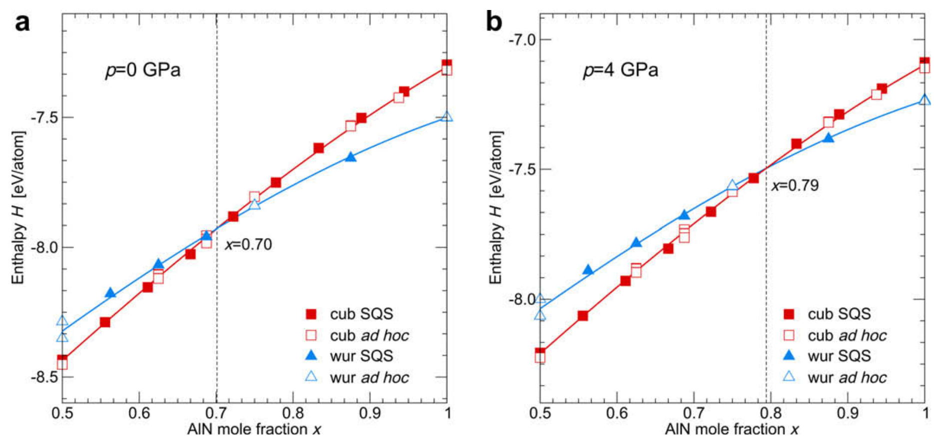

Figure 1. Enthalpies of the cubic (red squares) and wurtzite (blue triangles) phases of the $Ti_{1-x}Al_xN$ system (a) in equilibrium and (b) under a load of 4 GPa hydrostatic pressure. Full and open symbols represent the SQS and the ad hoc configurations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this paper.)

films [20] and thus this model consideration is based on a realistic situation. The dependence of x_{max} in $Ti_{1-x}Al_xN$ as a function of pressure is shown in Figure 2. The trend for increasing the solubility limit with applying larger compressive hydrostatic pressures is predictable since AlN transforms from the wurtzite to the cubic phase at ~16 GPa (a value based on both experiments and calculations) [30,31]. Since the cubic phase has a smaller volume per atom than the wurtzite phase [6], TiN maintains its cubic structure also under compression. Consequently, one would expect that above such pressure the cubic phase is stable for the whole compositional range: indeed, for ~14 GPa the enthalpies of the cubic and wurtzite phases crosses at $x_{max} = 1$, which is in agreement with the value for AlN [30,31].

Subsequently, we applied the same methodology to the $Cr_{1-x}Al_xN$ system. This system is more difficult when compared with $Ti_{1-x}Al_xN$ since the proper magnetic state of the Cr atoms must be taken into account. The difficulties with describing magnetic random alloys are responsible for the absolute values of the ab initio calculated x_{max} to be less accurate than in the case of $Ti_{1-x}Al_xN$.

The data of Mayrhofer et al. [9] used for the phase-stability calculation is based on ad hoc cells. To strengthen our analysis we the added SQS-based data-points of Rovere et al. [26]. In agreement with the $Ti_{1-x}Al_xN$ system, as for the $Cr_{1-x}Al_xN$ system the ad hoc data points fall into the vicinity of the SQS cells (see Figure 3a). The addition of compression increases the maximum AlN solubility of the cubic phase also in the $Cr_{1-x}Al_xN$ system. Under a compressive pressure of 4 GPa, x_{max} is around 0.79 (see Figure 3b) which is ~20% higher than x_{max} of ~0.66 for 0 GPa (see Figure 3a). The calculated maximum solubility of AlN in $Cr_{1-x}Al_xN$ as a function of pressure p is shown in Figure 4. As for the $Ti_{1-x}Al_xN$

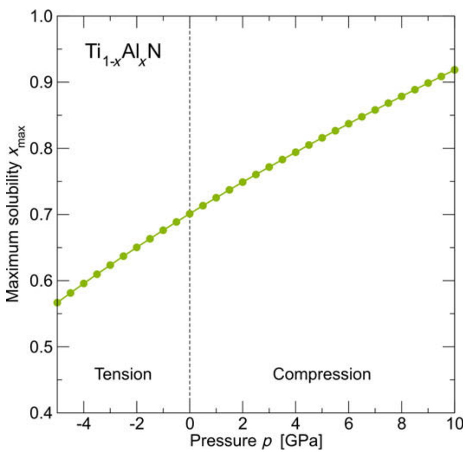

Figure 2. Calculated maximum mole fraction of AlN in the cubic B1 phase of the $Ti_{1-x}Al_xN$ system.

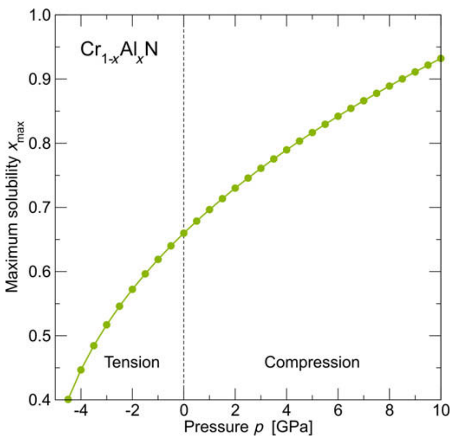

Figure 4. Calculated maximum mole fraction of AlN in the cubic B1 phase of the $Cr_{1-x}Al_xN$ system.

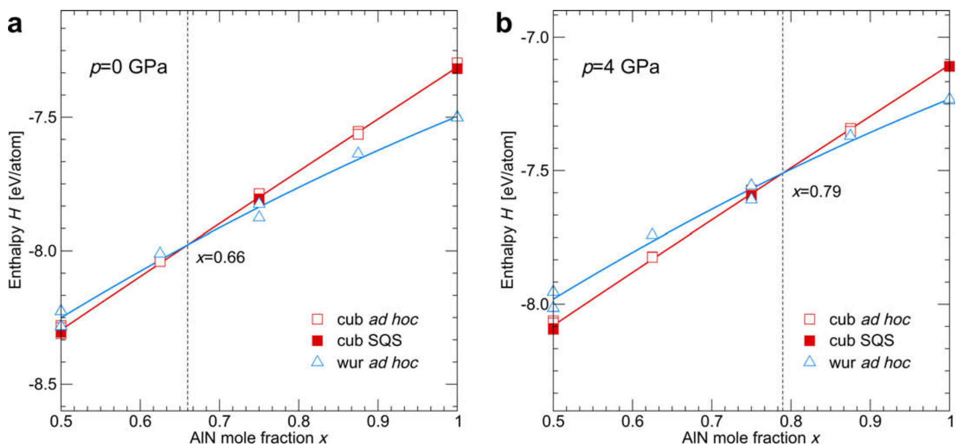

Figure 3. Enthalpies of the cubic (red squares) and wurtzite (blue triangles) phases of the $Cr_{1-x}Al_xN$ system (a) in equilibrium and (b) under a load of 4 GPa hydrostatic pressure. Full and open symbols represent the SQS and the ad hoc configurations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this paper.)

system, x_{max} increases with increasing compressive pressure. The dependence is steeper for $Cr_{1-x}Al_xN$ than in the case of $Ti_{1-x}Al_xN$, especially in the range of tensile hydrostatic pressures. Consequently, for tensile stresses in the coatings x_{max} drops rapidly below 0.5.

In conclusion, we have studied the effect of tensile and compressive hydrostatic pressure on the solubility limit x_{max} of AlN in the cubic B1 phase of $Ti_{1-x}Al_xN$ and $Cr_{1-x}Al_xN$ systems. It follows that x_{max} depends strongly on the applied pressure; compression of about 4 GPa increases x_{max} by ~0.1 AlN mole fraction for both systems. Although the studied stress state differs from the biaxial stress obtained experimentally in thin films during deposition, we can conclude that the built-in stresses influence the phase stabilities of these systems significantly.

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